



The Endo-lysosomal System in Parkinson's Disease: Expanding the Horizon

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Abstract

Parkinson's disease (PD) is the second most common neurodegenerative disorder after Alzheimer's disease, and its prevalence is increasing with age. A wealth of genetic evidence indicates that the endo-lysosomal system is a major pathway driving PD pathogenesis with a growing number of genes encoding endo-lysosomal proteins identified as risk factors for PD, making it a promising target for therapeutic intervention. However, detailed knowledge and understanding of the molecular mechanisms linking these genes to the disease are available for only a handful of them (e.g. *LRRK2*, *GBA1*, *VPS35*). Taking on the challenge of studying poorly characterized genes and proteins can be daunting, due to the limited availability of tools and knowledge from previous literature. This review aims at providing a valuable source of molecular and cellular insights into the biology of lesser-studied PD-linked endo-lysosomal genes, to help and encourage researchers in filling the knowledge gap around these less popular genetic players. Specific endo-lysosomal pathways discussed range from endocytosis, sorting, and vesicular trafficking to the regulation of membrane lipids of these membrane-bound organelles and the specific enzymatic activities they contain. We also provide perspectives on future challenges that the community needs to tackle and propose approaches to move forward in our understanding of these poorly studied endo-lysosomal genes. This will help harness their potential in designing innovative and efficient treatments to ultimately re-establish neuronal homeostasis in PD but also other diseases involving endo-lysosomal dysfunction.

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Introduction

Parkinson's disease (PD) is the second most common neurodegenerative disorder after Alzheimer's disease, and its prevalence is increasing with age. Approximately 10–15% of cases are familial, being linked to mutations in PD-related genes (e.g. *SNCA*, *VPS35*, *LRRK2*, *PRKN* (*Parkin*), *PINK1*, *PARK7* (*DJ-1*)),^{1,2} while the rest of cases are sporadic and influenced by individual genetic risk, environmental factors, or a combination thereof.² The recent advances in

sequencing technologies have allowed the analysis of cohorts of increasing size in Genome-Wide Association Studies (GWAS), leading to the identification of ever-increasing single nucleotide polymorphisms (SNPs) that are associated with PD risk.^{3–5} Even though we start seeing below the tip of the iceberg, explaining growing percentages of the genetic liability of PD, a large proportion of potentially significant risk factors remain to be identified and confirmed in diverse cohorts. In addition, synergistic effects of genetic variants with limited individual effects,⁶ or the existence of variants that affect the penetrance

of another one,⁷ highlight how genetic modifiers bring an additional layer of complexity that geneticists are beginning to explore.

Despite inherent challenges, GWAS and other genetic studies have provided extensive and crucial information to the community on genes and pathways driving PD pathogenesis. Notably, a clear enrichment for genes encoding endosomes and/or lysosomes localized proteins has been observed.^{4,8–10} As is the case for other age-related neurodegenerative proteinopathies, PD is tightly associated with a progressive decline in the ability of brain cells to degrade, discard or recycle defective cellular components such as depolarized mitochondria,^{11,12} misfolded/aggregated proteins,¹³ and lipids.^{14,15} Defects in intracellular vesicular trafficking can lead to improper sorting or delivery of cellular material targeted for degradation.^{16,17} Neurons and especially dopaminergic neurons of the *substantia nigra pars compacta* (SNpc) that are selectively vulnerable in PD, have a highly arborized architecture that renders them particularly dependent on adequate vesicular trafficking to either ensure delivery of cargoes targeted for degradation into lysosomes in the soma¹⁸ or targeting of soma-derived degradative lysosomes to distal axons.¹⁹ In addition, proper delivery or activity of factors and enzymes implicated in the maturation and degradative capacity of lysosomes are also key to ensuring the removal of potentially harmful material from the cells.^{20–22} Finally, failed degradation of cellular components will cause their accumulation, in turn affecting trafficking and the general degradative capacity of the cell.^{23–25} These mechanisms likely occur in both familial and sporadic forms of PD, although to varying degrees, and because of their complexity, it is not surprising that many different genetic variations can contribute to the disease.

Currently, we have a decent understanding of the molecular involvement in PD of only a few PD-linked endo-lysosomal genes: mostly those causing familial PD and the most common risk factors (*SNCA*, *VPS35*, *LRRK2*, *PRKN*, *PINK1*, *PARK7* (*DJ-1*)). However, this has not yet translated into disease-modifying therapy, and a better understanding of the lesser studied PD-linked genes in general (and endo-lysosomal genes in particular) appears essential to grasp the complexity of the mechanisms underlying PD pathogenesis. As in other fields of biomedical research, the study of these genes is lagging behind mainly due to historical bias.²⁶ For the purpose of this review, we focused on lesser-studied PD-associated endo-lysosomal protein-coding genes with a low number of research articles describing how they are implicated in disease at the molecular level. Notably, because some SNPs in PD-linked loci could affect several genes, we decided to be permissive and discuss not only the most likely candidate per locus but other endo-

lysosomal genes within these loci as well. Given the high number of lesser-studied genes in PD, and to maintain the focus on the endo-lysosomal system, we excluded genes encoding proteins strictly related to autophagy, endoplasmic reticulum or Golgi. The discussed genes, their molecular functions, and some of the associated genetic evidence are summarized in Table 1. This review aims at providing a valuable source of molecular and cellular insights into the biology of lesser studied PD-linked endo-lysosomal genes, at placing them in the cellular landscape of trafficking (Figure 1) and lysosomal function (Figure 2) and identifies key challenges that need to be taken on by the community to fill this knowledge gap. This will help and encourage researchers to harness the full depth of genetic clues and scrutinize these less popular genetic players.

Lesser studied genes of the endocytic and endosomal machinery in PD

Several GWAS studies have identified genes of the endocytic and endosomal systems as potential risk factors that can contribute to Parkinson's disease etiology. Some of the lesser-studied genes are discussed in this section, classified by functional categories.

Rab GTPases

The Ras-associated binding protein (Rab) family of small GTPases has recently emerged as a major player in PD pathology. Rabs maintain neuronal vesicular trafficking as well as synaptic architecture and integrity in the central nervous system.²⁷ About 60 Rabs have been discovered in eukaryotes pointing towards their key role in cellular physiology. Rab39B, which controls the anterograde transport of vesicles, is the most defined/studied risk gene in familial PD in the Rab family.^{27,28}

RAB29. GWAS studies identified risk variants in the *RAB29* gene (*PARK16* risk locus) in different ethnicities.²⁹ *RAB29* plays a role in the *LRRK2* pathway of PD pathogenesis. *LRRK2* (leucine-rich repeat kinase 2, *PARK8*) is one of the most common autosomal-dominant genetic causes linked to familial PD and has also been shown to have a causal effect in sporadic PD.^{30,31} Mutations in *LRRK2* accounts for 5–13% of familial PD and 1–5% of sporadic PD, which can vary depending on the population studied. In the North African Berber and Ashkenazi Jewish patient populations, this proportion rises to up to 40% of cases. The genetic association between *LRRK2* and PD is complex, with a lot of variation in penetrance depending on the specific familial mutations. As such, *LRRK2* can be considered as both a gene that causes familial PD, but also a modifier of the lifetime risk of devel-

Table 1 Summary of lesser-studied endo-lysosomal genes discussed, the molecular function of the protein they encode, and some of the genetic evidence linking them to PD. * OR = odds ratio; the 95% confidence interval is mentioned in parentheses. For other abbreviations, refer to abbreviations list.

Gene	Molecular function	PD association
Endocytic and endosomal machinery		
1. Rab GTPases		
i) <i>RAB29/</i> <i>PARK16</i>	functions to recruit LRRK2 to overloaded lysosomes and the trans-Golgi network (TGN), and stimulating its kinase activity	GWAS ²⁹ rs947211 OR = 0.91(0.88–0.94) GWAS ⁴³ rs823156 OR = 0.74 (0.68–0.81)
ii) <i>RAB25</i>	promotes the rerouting of integrins and receptor tyrosine kinases towards the plasma membrane through late endosomal recycling	GWAS ⁴³ chr1:154105678 OR = 1.73 (1.43–2.02) OR = 1.67 (1.41–1.98) ³⁰³ GWAS ³ rs34372695 OR = 0.66 (0.60–0.74)
2. ESCRT		
i) <i>VPS37B</i>	involved in the endocytic targeting of ubiquitinated protein to MVBs	GWAS ³⁰³ rs12817488 OR = 1.17 (1.09–1.25)
ii) <i>VPS25</i>	involved in regulating the endocytic targeting of ubiquitinated proteins to MVBs	GWAS rs144610244 ⁴³
iii) <i>CHMP2B</i>	mediates the recruitment of Vps4 and initiation of ESCRT complex dissociation	GWAS rs111338419 ⁴³
3. Inositol Phosphatases		
i) <i>INPP5F</i>	dephosphorylation of PI(4,5)P ₂ phosphoinositide to prevent its accumulation in the endosomal membrane	GWAS ³ rs117896735 OR=1.624 PD-GWAS gene prioritization WPPINA ¹⁰⁵
ii) <i>SYNJ1/</i> <i>PARK20</i>	a dual phosphatase with a Sac1 4-phosphatase domain and a 5-phosphatase domain that dephosphorylates PI(4,5)P ₂ phosphoinositide in early endosomes	c.773G > A, p.Arg258Gln genome-wide homozygosity mapping and exome sequencing ^{304,305}
4. Other vesicular trafficking genes		
i) <i>SH3GL2</i>	endocytic adaptor protein critical for clathrin-dependent and independent endocytosis	Two-stage meta-analysis GWAS ⁶⁶ rs13294100 OR = 0.92 (0.89–0.94)
ii) <i>TMEM230</i>	predominantly co-localizes with the TGN-enriched protein syntaxin 6 (STX6), and is involved in vesicle trafficking	Whole exome sequencing ⁷² WEG WGS ⁷⁴
iii) <i>SYT11</i>	shown to limit membrane invagination ensuring precision in vesicle retrieval	GWAS ³ rs34032320 OR = 0.52 (0.45–0.60)
iv) <i>GAK</i>	facilitates the clathrin uncoating of vesicles in trafficking by phosphorylating the coat adaptor component, AP2M1	GWAS ^{3,303} rs1564282 OR = 1.24 (1.17–1.31)
v) <i>SCAMP3</i>	involved in the regulation of morphology and composition of endosomes	PD-GWAS gene prioritization WPPINA ¹⁰⁵
Lysosomal machinery		
1. Ceramide metabolism		
i) <i>PSAP</i>	cleavage products give rise to the sphingolipid activator proteins/SAP (saposins) A, B, C, and D that regulate sphingolipid metabolism	PSAP c.1358A>C, p.Q453P Whole exome sequencing ¹²⁶
ii) <i>SCARB2</i>	encodes the lysosome membrane protein 2 (LIMP-2) that trafficks GCase from the ER through the Golgi and endosomes to the lysosomes	GWAS ^{3,4} rs6825004 OR = 0.92 (0.89–0.95)
iii) <i>ATP10B</i>	P4-type ATPase that acts as a lipid flippase in late-endo- lysosomes to translocate GlcCer and phosphatidylcholine (PC) from the luminal to the cytoplasmic leaflet of the late-endolysosomal membrane	Whole exome sequencing ¹⁴⁰
iv) <i>GALC</i>	encodes GalCase, an enzyme responsible for hydrolyzing GalCer and galactosylsphingosine GalSph	Two-stage meta-analysis GWAS ⁶⁶ rs8005172 OR = 1.08

(continued on next page)

Table 1 (continued)

Gene	Molecular function	PD association
v) <i>GLA</i>	encodes α -Gal A hydrolyzes terminal galactosyl residues from various complex glycoconjugated ceramides in lysosomes	Bi-directional sequencing ¹⁴⁸
vi) <i>ARSA</i>	encodes Arylsulfatase A, a lysosomal hydrolase that converts sulfatides, a major component of myelin, to galactosylceramides	Burden analysis SKAT-O test, meta-analysis ¹⁵⁴
vii) <i>ASAH1</i>	encodes Acid Ceramidase (aCerase) that hydrolyzes Cer to Sph and free fatty acid	Whole-exome sequencing ⁸ P = 0.031
viii) <i>SMPD1</i>	Acid sphingomyelinase (ASMase), encoded by <i>SMPD1</i> , converts sphingomyelin to Cer and phosphorylcholine	Whole-exome sequencing ⁸ P = 0.029
ix) <i>HEXA/B</i>	β -hexosaminidase (HEX) is responsible for the hydrolysis of the glycosphingolipid ganglioside GM2	Whole-exome sequencing ⁸
2. Cholesterol metabolism		
i) <i>NPC1/2</i>	Involved in the sensing and transport of cholesterol	Heterozygous mutations in NPC patients with Parkinsonism ¹⁷⁸ NPC1/2 genotyping in PD patients (not significant) ¹⁷⁷ Reviewed in Schneider et al. ¹⁸³
3. Glycosaminoglycan metabolism		
i) <i>NAGLU</i>	N-acetylglucosaminidase: hydrolyses the terminal N-acetyl-D-glucosamine residues in HS	GWAS ³ rs601999 OR = 0.92 (0.89–0.95)
ii) <i>IDUA</i>	iduronidase that hydrolyzes terminal unsulfated alpha-L-iduronosidic linkages in DS and HS promoting the degradation of these GAGs	GWAS ³ rs34311866 OR = 1.26 (1.22–1.31)
iii) <i>GUSB</i>	hydrolase β -glucuronidase, a homotetramer, which removes glucuronic acid from GAGs such as HS	GWAS ⁴
iv) <i>ARSB</i>	Arylsulfatase B hydrolyzes N-acetyl-D-galactosamine4-sulfate units in GAGs	WES LoF variant analysis rs138279020 ²⁰⁰
4. Sialic acid metabolism		
i) <i>NEU1</i>	Catalyzes the removal of sialic acid (Neu5Ac) residues from glycoproteins and glycolipids	GWAS ³ rs9267649 OR = 1.10 (1.06–1.15)
ii) <i>SLC17A5</i>	proton-coupled exporter of sialic acid (Neu5Ac) and other acidic hexoses from the lysosomes	Whole-exome sequencing ⁸ P = 0.005
5. Peptide metabolism		
i) <i>GRN</i>	Involved in lysosomal delivery and activation of GCase and other lysosomal enzymes	GWAS ³ rs5848 OR = 1.10 (1.05–1.15)
ii) Cathepsins		
a. <i>CTSB</i>	lysosomal protease of the papain family having endopeptidase and dipeptidyl carboxypeptidase function	Two-stage meta-analysis GWAS ⁶⁶ rs2740594 OR = 1.09 (1.07–1.12)
b. <i>CTSD</i>	lysosomal aspartic protease involved in the lysosomal recycling and degradation of many substrates, including α -syn	Whole-exome sequencing ⁸ P = 0.002
iii) <i>ABCB9</i>	transport polypeptides 6-59 amino acids long from the cytosolic space into the lysosomal lumen	GWAS ³⁰³ rs12817488 OR = 1.17 (1.09–1.25)
6. Ion Channels		
i) <i>TMEM175</i>	K ⁺ channel located in late endosomes and lysosomes helps maintain optimum pH	Two-stage meta-analysis GWAS ⁶⁶ rs34311866 OR = 1.23 (1.20–1.27)
ii) <i>ATP6V0A1</i>	a component of vacuolar ATPase (V-ATPase), a multisubunit proton pump that mediates the acidification of organelles	GWAS ³ rs34779517 OR = 0.91 (0.88–0.94) Two-stage meta-analysis GWAS ⁶⁶ rs601999 OR = NA
iii) <i>ATP6V1D</i>	contains the ATP catalytic site for the transport of H ⁺ to mediate the acidification of endo-lysosomes	GWAS ⁴³ rs10144002
Other endo-lysosomal genes		

Table 1 (continued)

Gene	Molecular function	PD association
1. <i>VAMP4</i>	recruitment of clathrin adaptor proteins; TGN-from/to-endosome vesicle transport	GWAS ⁴ rs11578699 OR = 0.93 (0.91–0.95)
2. <i>VT11B</i>	part of a protein complex that is required for late endosome-lysosome membrane fusion	GWAS ³ rs10913565 OR = 0.91 (0.88–0.95)
3. <i>SPPL2B</i>	member of the GXGD family of aspartic proteases mediate the shedding of several substrates all belonging to type 2 transmembrane protein subfamily	WGS-based imputation and analysis of existing exome genotype data ³⁰⁶ rs62120679 OR = 1.097
4. <i>ZFYVE26</i>	SPASTIZIN hypothesized to recruit Beclin 1 and other partners involved in autophagosome maturation	GWAS ³ rs2057119 OR = 0.96 (0.92–0.99)
5. <i>ATP11B</i>	catalytic component of a P4-ATPase flippase complex that transports aminophospholipids, PS, and PE, from the outer to the inner leaflet of intracellular membranes	GWAS ³
6. <i>CLCN3</i>	encodes the CIC3 vesicular 2Cl [−] /H ⁺ exchanger that maintains the ion homeostasis and acidity in the endosomal lumen	GWAS ⁴ rs62333164 OR = 0.94 (0.92–0.96)

oping PD. Point mutations (G2019S, I2020T, R1441C/G/H, Y1699C, and N1437H) in the LRRK2 functional enzymatic domains result in autosomal dominant PD. Some rare LRRK2 point mutations such as I2020T appear to be highly penetrant, whereas the common G2019S LRRK2 mutation is just moderately penetrant. LRRK2 missense variants G2385R and R1628P are associated with increased PD risk in Han Chinese populations, but other haplotypes (e.g. N551K-R1398H) present with lower sporadic PD risk. The presence of the G2019S mutation in the kinase domain of LRRK2 has been demonstrated to give rise to a hyperactive kinase displaying increased autophosphorylation and substrate phosphorylation.^{32–34}

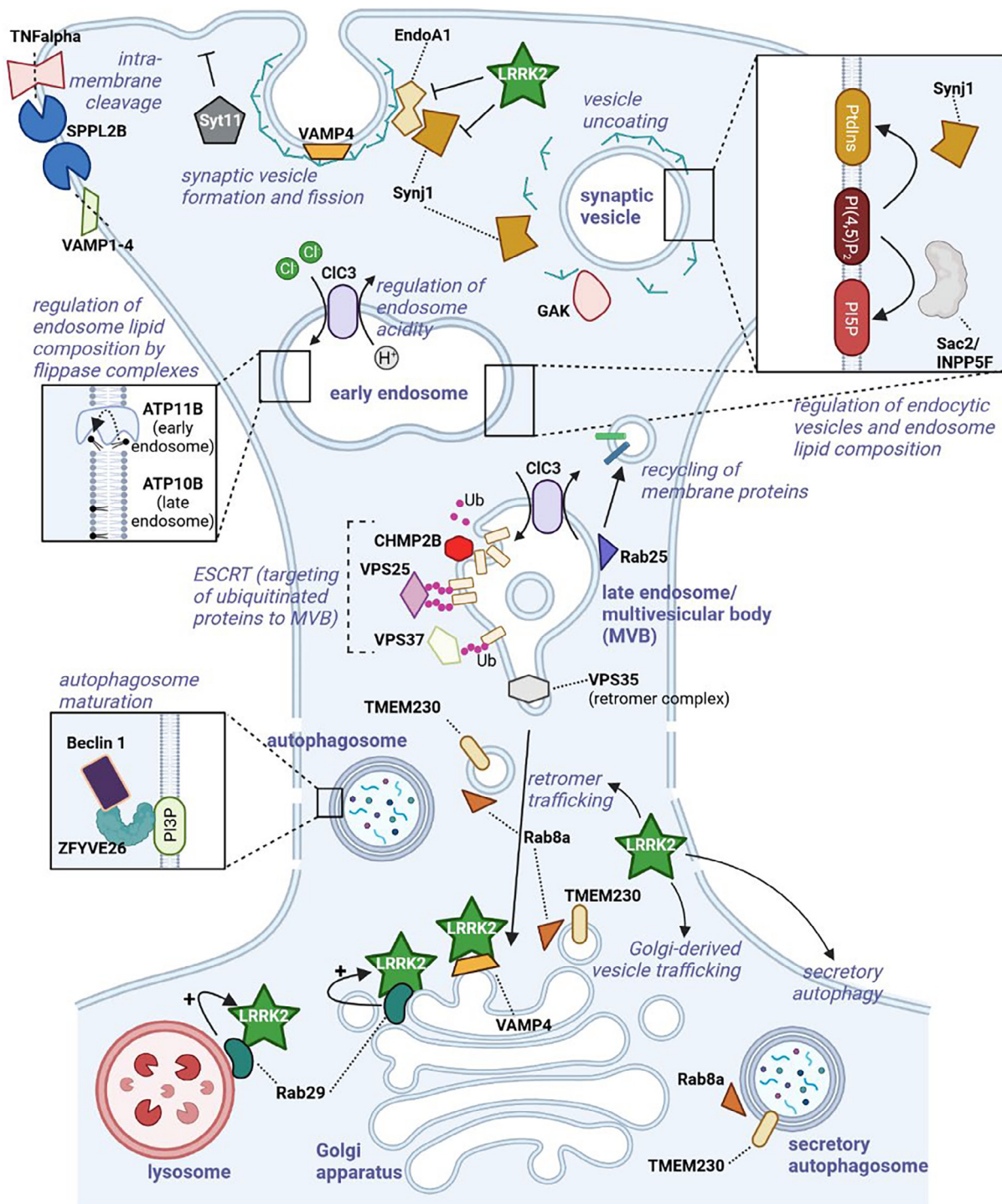
LRRK2 phosphorylates most Rabs, including Rab29. However, Rab29 also functions upstream of LRRK2 by binding to its N-terminus ankyrin domain, recruiting it to overloaded lysosomes and the trans-Golgi network (TGN), and stimulating its kinase activity.^{27–28,35} Rab29 was shown to have a poor affinity to GTP and its GTPase activity did not affect LRRK2 binding suggesting that its regulation on LRRK2 activity functions differently from other Rabs. Pathogenic LRRK2 (R1441G/C and Y1699C mutants) are more readily recruited to the Golgi by Rab29.³⁶ *Rab29* KO mice displayed a similar histological phenotype as *LRRK2* KO mice, with an accumulation of secondary lysosomes in the proximal tubules of the kidney.³⁷ Rab29[T71A, S72A] non-phosphorylatable mutant did not affect LRRK2 activation but Rab29[T71E, S72E] phosphomimetic abolished its ability to activate LRRK2, which may be due to its inactivation. This mimics a negative feedback loop where LRRK2-mediated Rab29 phosphorylation prevents Rab29 from activating LRRK2.³⁶ While Rab29's residue Ser72 was suggested to influence TGN morphology,³⁸

Thr71 was shown to be the major target residue of LRRK2 kinase activity when in a GTP-bound state.³⁵ Recent work by the Holzbaur team elucidated the mechanism of physiological dysfunction in neurons by Rab29/LRRK2. The LRRK2 G2019S hyperactive mutant or the Rab29-induced LRRK2 hyperactivity was shown to disrupt the retrograde axonal transport of autophagosomes by creating an imbalance in motor protein activity. This resulting accumulation of autophagosomes could explain neuronal death in PD.^{39,40} Since a role for Rab29 as the master regulator of LRRK2 has been defined (which was recently challenged by Karogelopolou et al.,⁴¹ targeting the activity of Rab29 might be a therapeutic option to treat LRRK2 autosomal dominant cohort of PD-affected patients and sporadic patients with hyperactive LRRK2. Interestingly, another recent study showed that RAB29 positively modulates the pathogenic G2019S LRRK2-mediated non-Golgi-related trafficking deficits, highlighting the possibility that RAB29 may play additional physiological roles important for proper endo-lysosomal trafficking unrelated to its proposed role as a LRRK2 activator at the Golgi.⁴² Further studies in animal models need to be undertaken to authenticate Rab29 as a potential target of intervention.

RAB25. Meta-analysis of genetic association studies in PD patients have shown a significant association of SNPs from the *SYT11-RAB25* locus with PD risk.⁴³ Especially, *RAB25* is the closest gene to the significant SNP rs34372695 (odds ratio = 0.66; 95% confidence interval = 0.60–0.74),³ being located just 914 bp downstream of this SNP.⁴³ Whether and how that and other SNPs affect RAB25 expression and/or function remains to be determined.

Together with Rab11a, and Rab11b, Rab25 is a member of the Rab11 subfamily. These Rabs regulate the recycling of plasma membrane-associated proteins following their internalization. Rab25's GTP binding domain carries a novel sequence WDTAGLE, that differs from the Rab family consensus WDTAGQE sequence by a Q to L substitution.⁴⁴ Rab25 is localized in the membrane-associated recycling vesicular structures of neurons that are positive for VAMP-2, H⁺/

K⁺ ATPase, and secretory carrier proteins (SCAMPs).⁴⁵ By promoting the rerouting of integrins and receptor tyrosine kinases towards the plasma membrane through late endosomal recycling, Rab25 prevents the lysosomal degradation of proteins. This function of Rab25 was shown to depend on the presence of the late endosomal and lysosomal factor CLIC3.⁴⁵ The function of Rab25 in PD still remains unclear. Strategies that modulate Rab25-dependent protein–protein inter-



actions and function will be key to understanding the role of this protein.

ESCRT

The molecular machinery known as Endosomal Complexes Required for Transport or ESCRT recognizes ubiquitinated proteins in early endosomes and multivesicular bodies (MVBs). The proteins belonging to this family were termed Vps (vacuolar protein sorting). The ESCRT system comprises four different protein complexes, namely ESCRT-0, -I, -II, and -III, and the Vps4 ATPase complex. ESCRT-0 first recognizes the ubiquitinated proteins and the other complexes are sequentially recruited to the endosomal membrane via protein–protein interactions. While ESCRT-I and -II further concentrate ubiquitinated proteins, ESCRT-III along with Vps4 ATPase complex participates in the budding and abscission of intraluminal vesicles (ILVs) of MVBs from the endosomal membrane thereby completing MVB biogenesis. The ESCRTs are recycled into the cytoplasm by the action of ATPase Vps4 and co-factor Vta1.^{46–}

⁴⁷ In the context of PD, the uptake of α -syn aggregates in B103 neuroblastoma cell line is at least in part by clathrin-mediated endocytosis. These aggregates are subject to ESCRT-mediated trafficking in MVBs and degradation in lysosomes.⁴⁸

VPS37B. Vps37 is one of the four subunits of the ESCRT-I complex. Vps37 has four isoforms (A, B, C, and D). Vps37b positively regulates the viral budding via host ESCRT machinery. It is involved in the endocytic targeting of ubiquitinated protein to MVBs.⁴⁹ It was shown that Vps37b could recruit

TSG101/ESCRT-I activity and promote the budding of HIV-viral particles from Cos-7 cells.⁴⁹ We propose the hypothesis that this protein is involved in the egress of α -syn from neurons promoting trans-cell propagation of α -syn aggregates in PD affected brains, similar to its involvement in viral propagation. A recent study showed that the pharmacological or genetic inhibition of p38MAPK in primary cortical neurons expressing p25 α and α -syn induces ESCRTI-dependent degradation of α -syn. This alleviated accumulation of α -syn and autophagosomes, decreased synucleinopathic toxicity, while lowering the amount of secreted α -syn in a TSG101-dependent manner.⁵⁰

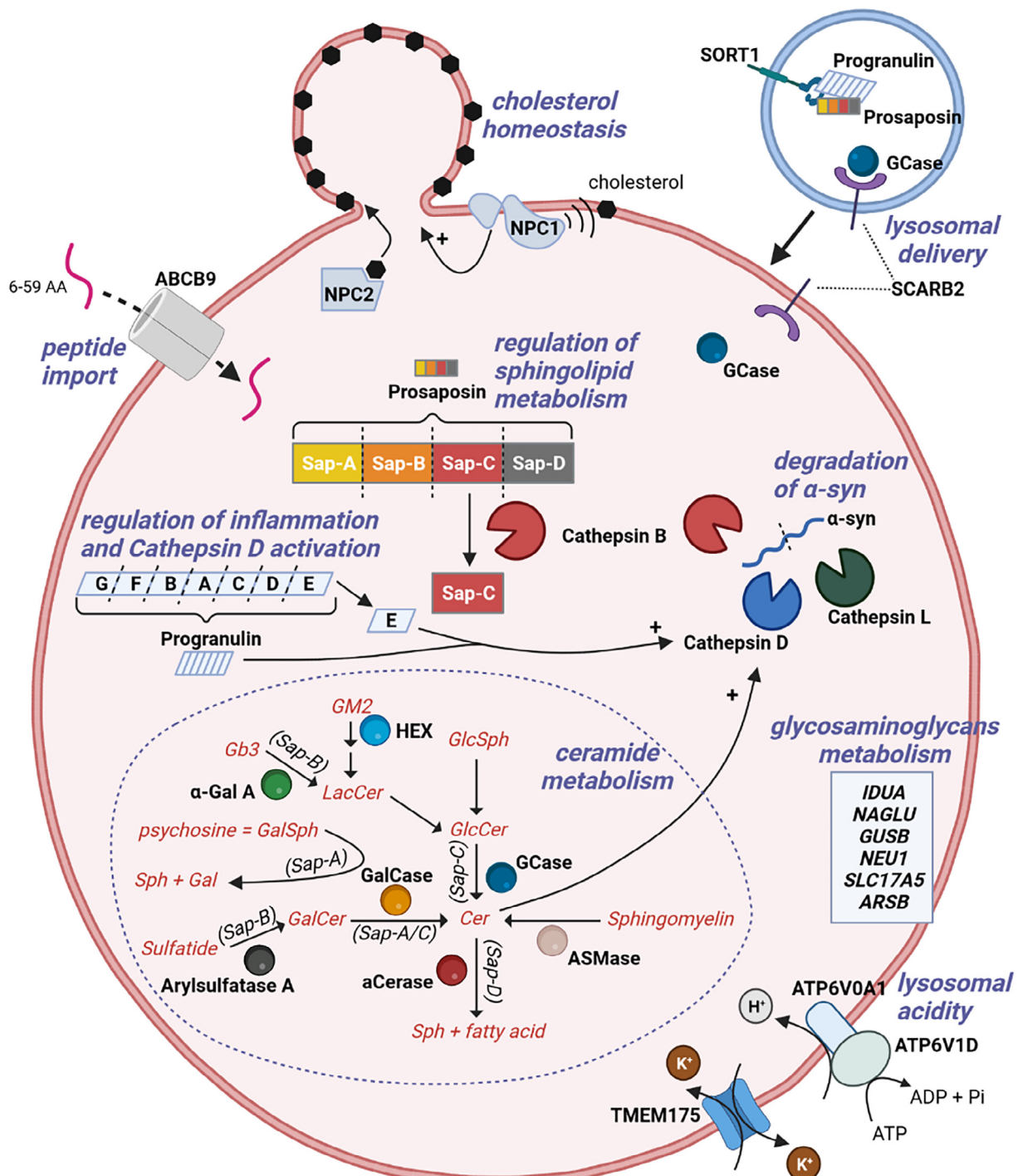
VPS25. Belonging to the ESCRT-II complex, Vps25 (also termed EAP20) appears as a dimer in the heterotetrameric complex. Vps25 is also involved in regulating the endocytic targeting of ubiquitinated proteins to MVBs. Deletion of *Vps25* resulted in the complete blockade of Avian sarcoma virus budding from 293/E cells.⁵¹ Similar to Vps37b, Vps25 could also be hypothesized to be involved in α -syn release and propagation.

CHMP2B. Belonging to the ESCRT-III complex, CHMP2 has two isoforms: A and B. CHMP2 mediates the recruitment of Vps4 and initiation of ESCRT complex dissociation. In neurons, it is associated postsynaptically with the cytoskeleton and is thought to strengthen the dendritic spines.⁵² *CHMP2B* mutations are reported to cause autosomal dominant frontal temporal dementia (FTD) in humans.⁵³ Mutated *CHMP2B* causes the death of cortical neurons in culture as a result of dendritic retraction, and accumulation of ubiquitinated proteins and autophagosomes.⁵⁴ Several immunohis-

Figure 1. PD-linked, understudied players of endocytic and secretory trafficking pathways. Synaptic vesicle (SV) formation and fission: Syt11 inhibits endocytosis and vesicle replenishment by limiting membrane invagination. VAMP4 recruits clathrin adaptor proteins, while EndoA1 induces membrane curvature and recruits downstream factors required for fission and/or uncoating such as Synj1. The latter dephosphorylates PI(4,5)P₂ to facilitate the uncoating of SV. GAK also facilitates the uncoating of clathrin by phosphorylating a coat adaptor component. Endosomes homeostasis: The endosomes membrane lipid composition, which is crucial to maintain the identity and functionality of this membrane, is regulated by flippase complexes that contain the ATP11B (early endosomes, EE) or ATP10B (late endosomes, LE) subunits. PIP phosphatases Synj1 and Sac2 also contribute to this regulation. The CIC3 2Cl[−]/H⁺ exchanger regulates the acidity of the endosomal compartment. Sorting in late endosome: Rab25 regulates the recycling of internalized membrane-associated proteins to the plasma membrane. The ESCRT complexes components VPS37, VPS25, and CHMP2B facilitate the targeting of ubiquitinated proteins to multivesicular bodies (MVB). The retromer complex component VPS35 regulates retrograde trafficking from endosomes to Golgi, and TMEM230 may participate in that process together with Rab8a and LRRK2. Golgi trafficking and secretory autophagy: TMEM230, together with Rab8a and LRRK2, converge on regulating Golgi-derived vesicle trafficking and secretory autophagy. The Golgi v-SNARE VAMP4 interacts with LRRK2 to regulate membrane fusion during retrograde trafficking from endosomes to Golgi. Rab29 can recruit and stimulate the LRRK2 kinase at the trans-Golgi network. Lysosome homeostasis: Rab29 can also recruit and stimulate LRRK2 at the surface of overloaded lysosomes, facilitating the release of lysosomal content into the cytosolic space. Autophagosome maturation: ZFYVE26 (SPASTIZIN) induces autophagosome maturation by binding to PI3P at the surface of forming autophagosomes, and recruiting Beclin 1 and other maturation factors. Intramembrane cleavage: Neurons enriched SPPL2B performs intramembrane cleavage of several substrates, including TNF- α . It also cleaves vesicular trafficking regulators VAMP1, 2, 3, and 4, regulating their turnover.

tochemical studies on patient brains have reported CHMP2B immunoreactivity in intracytoplasmic and axonal Lewy bodies in α -synucleinopathies including PD and dementia with Lewy bodies (DLB).^{55,56} Overexpression or knock-down of CHMP2B in neuronal B103 cells decreased or increased the accumulation of α -syn oligomers respectively.^{57,58} It was also reported that α -syn aggregates disrupt the ESCRT-III complex causing α -syn exocytosis resulting in cell-cell propagation.⁴⁸ Overexpression

of CHMP2B in the hippocampus caused rapid clearance of α -syn aggregates and amelioration of neurodegenerative pathology in a mouse model overexpressing human α -syn from the murine Thy1 promoter.⁴⁸ On the contrary, siRNA-induced depletion of CHMP2B increased exocytosis and intercellular propagation of α -syn aggregates in B103 neuroblastoma cell lines and primary mouse cortical neurons.^{47,48}



Inositol phosphatases

Inositol phosphatases belong to the family of phosphodiesterases and are involved in the phosphatidylinositol signaling pathway. Phosphatidylinositol phosphates (PIPs) are phosphorylated derivatives of the membrane phospholipid, phosphatidylinositol (PI) which is synthesized at the endoplasmic reticulum from the precursor glucose-6-phosphate. The subcellular membranes of different cell organelles have varying PIP compositions providing a unique lipid signature that aids in targeting the delivery of proteins to specific organelles and facilitates cellular processes. PI makes up to 10–20% of the total mammalian membrane, but the total levels of PIPs are maintained at very low levels (approximately 2–5% of PI) suggesting a high regulation in synthesis, maintenance, and turnover of PIPs resulting in a fine-tuning of PI signaling.⁵⁸

INPP5F. *INPP5F* was recently identified as a risk locus in PD by the GWAS study conducted by Nalls et al.⁴ *INPP5F* encodes Sac2, a member of the Sac family of inositol phosphatases. Sac2 is reported to act as a PI4 phosphatase. It colocalizes with the early endosomes and aids in the dephosphorylation of PI(4,5)P₂ phosphoinositide to prevent its accumulation in the endosomal membrane.⁵⁹ Sac2 was shown to be selectively recruited to endocytic cargoes (e.g. transferrin) that are programmed for recycling.⁶⁰ *INPP5F* mRNA is highly enriched in the brain, while Sac2 protein interacts with Rab4 and Rab5 and is preferentially expressed in neurons.^{60,61} (transferrin) that are programmed for recycling.

clinging.⁶⁰ Sac2 mRNA is highly enriched in the brain, while Sac2 protein interacts with Rab4 and Rab5 and is preferentially expressed in neurons.^{60,61} Sac2 was recently identified as a risk locus in PD by the GWAS study conducted by Nalls et al.⁴

SYNJ1. Synaptojanin1 (Synj1, *PARK20*) is a dual phosphatase with a Sac1 4-phosphatase domain and a 5-phosphatase domain that dephosphorylate PI(4,5)P₂ phosphoinositide in early endosomes. Synj1 has a highly variable C-terminal proline-rich domain for endoA1-mediated recruitment in SV endocytosis.⁶² The most established function of Synj1 is in the clathrin uncoating of synaptic vesicles.⁵⁸ The R258Q inactivating homozygous missense mutation in the highly conserved Sac inositol phosphatase domain of Synj1 causes early-onset Parkinsonism (EOP) with epilepsy.^{61,63} A differentially expressed gene analysis study also points towards Synj1 as a risk factor for PD.⁶⁴ Synaptojanin1 (Synj1) is a dual phosphatase with a Sac1 4-phosphatase domain and a 5-phosphatase domain that dephosphorylate PI(4,5)P₂ phosphoinositide in early endosomes. Synj1 has a highly variable C-terminal proline-rich domain for endoA1-mediated recruitment in SV endocytosis.⁶² The most established function of Synj1 is in the clathrin uncoating of synaptic vesicles.⁵⁸ The R258Q inactivating homozygous missense mutation in the highly conserved Sac inositol phosphatase domain of Synj1 causes early-onset Parkinsonism (EOP) with epilepsy.^{61,63} A differentially expressed gene analysis study also points towards Synj1 as a risk factor for PD.⁶⁴

Figure 2. Understudied lysosomal proteins with a potential genetic link to PD. Cholesterol homeostasis: The transmembrane protein NPC1 senses cholesterol in the lysosomal membrane (and other membranes of the endolysosomal system), inducing budding for the ultimate delivery of cholesterol in target membranes. NPC2 transports cholesterol from the donor to acceptor membranes. Lysosomal delivery: SCARB2 mediates the trafficking of GCase through the secretory pathway for its delivery to the lysosomal lumen. Interaction between Prosaposin and Progranulin regulates their delivery to lysosomes by the sortilin SORT1. The ABCB9 transporter mediates the translocation of 6–59 amino acid long peptides from the cytosol to the lysosomal lumen for degradation. Degradation of α -synuclein (α -syn): Cathepsins B, D, and L were shown to cleave α -syn, facilitating its clearance. Cathepsin D activity is positively and directly regulated by Ceramide, progranulin, and granulin E. Regulation of sphingolipid metabolism by Saposins (Sap-A/B/C/D): Prosaposin matures by proteolytic cleavage in the lysosomes. Cathepsin B favors Sap-C maturation. Sap-A, Sap-B, Sap-C, and Sap-D act by facilitating the activity of various enzymes related to sphingolipid metabolism, including ceramide metabolism. Ceramide metabolism: A clear enrichment of genes encoding enzymes of this pathway are found as risk factors for PD, including GCase (glucosylceramidase), GalCase (galactosylceramidase), ASMase (acid sphingomyelinase), α -Gal A (α -galactosidase A), aCerase (acid ceramidase), Arylsulfatase A, and HEX (β -hexosaminidase). Glycosaminoglycans (GAGs) metabolism: several genes linked to PD are regulating GAGs metabolism in the lysosome. Their link to PD is unclear but it can be hypothesized that decreased GAGs degradation leads to reduced degradation of α -syn and its subsequent accumulation. Lysosomal acidity: Acting as a K⁺ “leak” channel, TMEM175 influences lysosomal pH (increased acidity upon TMEM175 deficiency). The two v-ATPase subunits, ATP6V0A1 and ATP6V1D, are part of the enzymatic complex responsible for pumping protons into the lysosomal lumen. Miscellaneous: Progranulin and granulin peptides display anti- and pro-inflammatory properties respectively. This may not be via a lysosome-specific action, but rather following secretion in the extracellular space. Gb3: globotriaosylceramide; GM2: GM2 ganglioside; LacCer: lactosylceramide; GalSph: galactosylsphingosine; Sph: sphingosine; Gal: galactose; GalCer: galactosylceramide; GlcSph: glucosylsphingosine; GlcCer: glucosylceramide; Cer: ceramide; AA: amino acids.

Sac2 and Synj1's Sac domains have overlapping functions which suggest a compensatory effect in the absence of one of the two proteins. This is supported by findings that inactivation of Sac2 in Synj1 R258Q PD mutants accelerates the occurrence of neurodegeneration due to the accumulation of endocytic factors in the axon terminals of dopaminergic neurons in culture and *in situ* in brains sections from mice models.⁶¹ Synj1 was shown to be a direct substrate of LRRK2 phosphorylation *in vitro* in mid-brain neurons, and phosphorylation in its proline-rich domain resulted in the disruption of endophilin-Synj1 interaction required for SV endocytosis.⁶⁵ LRRK2, endoA1, and Synj1 are suggested to form a tripartite signaling complex that controls synaptic membrane trafficking and autophagy, both shown to be disrupted in PD.

Other vesicular trafficking genes

SH3GL2. *SH3GL2*, encoding endoA1, was recently identified in a meta-analysis of GWAS studies as a novel risk factor for PD.⁶⁶ Increased expression of endoA1 was causally linked to PD by a recent GWAS study conducted by Bandres-Ciga and colleagues.⁶⁷ Located in Chr 9, the risk loci rs10756899 near the *SH3GL2* gene was the top SNP out of the 15 SNPs in the region as reported by Nalls et al.⁴ Mammalian endophilin A consists of a family of three homologous proteins encoded by the genes *SH3GL2* (endoA1), *SH3GL1* (endoA2), and *SH3GL3* (endoA3). EndoA1 is the only brain-specific isoform and a well-characterized endocytic adaptor protein critical for clathrin-dependent and independent endocytosis.⁶² It coordinates synaptic vesicle (SV) endocytosis⁶⁵ and exocytosis⁶⁸ at nerve terminals. It contains two domains: the N-terminal BAR allows detection and induction of membrane curvature, while the C-terminal SH-3 domain acts as a protein hub, mediating protein interactions to orchestrate the functions of factors that remodel the membrane and perform constriction and scission.⁶⁸

Lack of functional endophilins was shown to cause accumulations of clathrin-coated vesicles at the synaptic terminals of both invertebrate and vertebrate models.⁶⁹ Deletion studies on endoA1 exhibited age-related motor impairments and progressive ataxia in a cerebellar ataxia mice model.⁷⁰ EndoA1 was shown to be a direct substrate of the LRRK2 kinase in the neuromuscular junction of *drosophila*.⁷¹ Studies suggest that the LRRK2 G2019S mutation disrupts SV recycling in the mid-brain DA neurons of *SYNJ*^{+/-} mice by phosphorylating Synj1 and compromising its interaction with endoA1.⁶⁵ Synj1 being a close binding partner of endoA1,⁶² places endoA1 in a common pathway of PD pathogenesis where *LRRK2* and *SYNJ1* signaling converge.

TMEM230. Four mutations in *TMEM230* were reported by Deng et al.⁷² and suggested to be

autosomal-dominant. Only a single missense variant (c.422G > T, p.R141L) was confirmed to co-segregate with the disease.⁷³ Whole-exome sequencing and whole-genome sequencing analysis of *TMEM* protein family genes also identified linkage of three rare variants of *TMEM230* with PD.⁷⁴ However, several studies in different cohorts by other groups failed to identify the PD-linked mutations in *TMEM230*.^{73,75} One study also rejects the possibility of *TMEM230* as a PD risk gene.⁷⁶ This could be explained by the presence of a large number of polymorphisms in the *TMEM230* gene observed across various ethnicities and cohorts which could pose a challenge to a conclusive finding. The exact reasons for these discordant findings will need to be tackled by the PD genetics community. Here, we decided to discuss this gene regardless of the ongoing debate but encourage readers to pay attention to its resolution in the future.

Studies have reported that Transmembrane Protein 230 (also termed C20orf30) is involved in vesicle trafficking (secretion/recycling of vesicles including synaptic vesicles in neurons), Golgi secretion, endosomal recycling, autophagy, and neurotoxicity. *TMEM230* predominantly co-localizes with the TGN-enriched protein syntaxin 6 (STX6) but is also observed to localize with other vesicular proteins like VPS35, VMAT2, Rab5a, and Rab11a.^{1,72,73}

TMEM230 distribution was found to be confined to vesicular membrane structures in SH-SY5Y cells and mouse dopaminergic (DA) neurons of the SNpc.⁷² Among the two isoforms encoded, the shorter isoform-2 accounts for 95% of the total protein and harbors a highly conserved amino acid sequence that includes its two transmembrane segments (TMS) as well as N- and C- termini exposed to the cytosol. *TMEM230* mRNA is highly expressed in the neural tissues of most mammals, mainly detected in the midbrain, neocortex, and spinal cord. *TMEM230* has no reported sequence homology to any known protein.^{1,72,73} More than half of the reported mutations in *TMEM230* are found in and around the TMSs. There is a growing body of evidence that *TMEM230* shares pathogenic pathways with vesicular trafficking proteins encoded in PD-causing genes like α -syn, LRRK2/Rab8a, Vps35, and Pink1 suggesting its role in PD-related neurodegeneration.^{72,77} *TMEM230* was observed to be consistently colocalized with SNCA-positive Lewy bodies and Lewy neurites.⁷² The function of the LRRK2 and Pink1 substrate Rab8 (involved in vesicle trafficking) was impaired when *TMEM230* was mutated in HEK293-FT cells suggesting a shared pathway for *TMEM230* and Rab8a in the development of PD and Lewy pathology.⁷⁷⁻⁷⁹ A partial co-localization of *TMEM230* with Vps35 (encoded by the *VSP35* gene, in which mutations cause autosomal dominant late-onset PD) was observed, suggesting they could act together in regulating retromer trafficking function.⁷²

Further studies on TMEM230, using genetic animal models for instance, need to be undertaken to confirm its role in PD incidence and neurodegeneration.

SYT11. Synaptotagmins (Syts) are secretory proteins composed of a short N-terminus, a transmembrane domain, and an evolutionarily conserved Ca^{2+} binding cytoplasmic tandem C2 domain.^{80,81} Syts are major components of the vesicle trafficking machinery. Syt11 is a Ca^{2+} -insensitive form (due to an aspartate-serine substitution in its C2A domain) that has recently been implicated as a novel risk factor in PD.⁸² Studies reveal a function for Syt11 in a non-conventional vesicular pathway ensuring precision in vesicle retrieval and hence deemed essential for normal development, synaptic function, and memory formation.^{80,81} Syt11 is a substrate of Parkin (*PRKN* gene, an E3 ubiquitin ligase in which loss-of-function mutations cause autosomal recessive PD). It was shown that Syt11 was an essential component in mediating Parkin-linked neurotoxicity.⁸³ Syt11 accumulation in DA neurons leads to late-onset neuronal loss and behavioral deficits in Syt11 overexpressing mice.⁸³ In the dorsal root ganglion and hippocampal neurons, Syt11 was shown to limit membrane invagination. This resulted in endocytosis inhibition and reduced vesicle replenishment.⁸¹ Syt11 overexpression in mice leads to depression of DA transmission and subsequent and progressive loss of DA neurons of the SNpc. Syt11 thus appears as an early player in genetic PD cases.⁸³

Syt11 was also shown to interact (split-ubiquitin yeast two-hybrid⁸⁴ and be regulated by *ATP13A2* (previously *PARK9*), which is mutated in autosomal recessive early-onset PD.⁸⁴ *ATP13A2* encodes a P5B-type ATPase, an integral 10-transmembrane spanning ATPase whose function was recently elucidated to be a late endo-lysosomal polyamine transporter.⁸⁵ Downregulation of *ATP13A2* expression levels or loss-of-function of *ATP13A2* resulted in lysosomal dysregulation characterized by impaired substrate degradation and autophagosome clearance in DA neurons, while also leading to α -syn misfolding and accumulation⁸⁴. The polyamines were shown to induce cytotoxicity which was exacerbated by lysosomal dysfunction due to *ATP13A2* loss.⁸⁵ Deficiency of *ATP13A2* and familial *ATP13A2* mutations were demonstrated to induce α -syn aggregation, membrane association and multimerization in primary cortical neurons and SH-SY5Y neuroblastoma cells.^{84,86} This has however been challenged. A study in *ATP13A2* KO mice found no α -syn positive Lewy bodies even in aged animals, and this KO did not exacerbate neurodegeneration nor α -syn insolubility when combined to α -syn A53T overexpression.⁸⁷ α -syn aggregation was also not found in the post-mortem brain of an *ATP13A2*-linked Kufor-Rakeb patient with Parkinsonism.⁸⁸

Bento et al. revealed a positive regulatory role for *ATP13A2* on Syt11 expression levels: *ATP13A2* favors Syt11 expression via the TSC2/mTORC1/TFEB transcriptional axis, while it decreases Syt11 ubiquitination and subsequent proteasomal-dependent degradation. In HeLa cells, perturbations in the *ATP13A2*-Syt11 pathway were linked to PD-relevant pathogenic events, including dysfunction of lysosomes, autophagy, and α -syn A53T accumulation.⁸⁹ Although there are contradicting claims about *ATP13A2* and α -syn aggregation, the leading hypothesis is that the autophagy dysfunction caused by *ATP13A2* loss causes the abnormal accumulation of α -syn.^{84,86,89,90}

GAK. The chromosome 4p16.3 region is one of the strongest PD risk loci and comprises several candidate genes including *GAK*, *TMEM175*, *DGKQ*, *SLC26A1*, and *IDUA*.^{91,92} The risk allele of the GWAS SNP rs1564282 in the *GAK* gene showed a marked association with susceptibility to familial and sporadic PD in Chinese, Taiwanese, and Caucasian populations.^{92–96} rs1564282 was associated with an increase in the expression of several genes including *SNCA* in SNpc of PD patients.^{92,93}

Cyclin G-associated kinase (GAK) (also known as auxilin 2, a member of the Numb-associated kinase (NAK) family) is a ubiquitously expressed serine/threonine kinase. By phosphorylating a coat adaptor component, AP2M1, it facilitates the clathrin uncoating of vesicles in trafficking. GAK was recently reported to have a role in α -syn expression and toxicity. GAK was shown to be required for CTSD maturation,^{97–99} the main lysosomal enzyme involved in α -syn degradation,¹⁰⁰ but its involvement in the trafficking of CTSD from TGN to lysosomes is controversial.^{97–99} GAK knockdown in α -syn over-expressing HEK293 cells increased cellular toxicity and total α -syn protein levels.⁹³ GAK was shown to form a complex with the well-studied PD-risk protein kinase LRRK2, which has been implicated in the trafficking of Golgi-derived vesicles to the lysosomal-autophagy system.¹⁰¹ GAK was also shown to have a co-chaperone function with Hsp70 for clathrin uncoating during endocytosis in non-neuronal and neuronal cells.^{102–104}

SCAMP3. SCAMP3 was reported to be a novel candidate PD risk gene by a protein interactome-based approach on candidate genes from GWAS risk loci.¹⁰⁵ SCAMP3 is an integral membrane protein belonging to the carrier membrane protein (SCAMP) family. This family consist of 5 isoforms; SCAMPs 1–3 being the longer isoforms, while SCAMP 4 and 5 are shorter. SCAMPs are involved in the intracellular trafficking of diverse transporters and receptors, via their interaction with factors implicated in protein sorting and vesicle formation.¹⁰⁶

SCAMP3 was recently shown to be a novel player in the regulation of morphology and composition of endosomes. It has also been shown to modulate EGFR turnover by ubiquitination and interaction with ESCRT components.¹⁰⁶ No molecular role for SCAMP3 in PD has been described so far.

Lesser studied lysosomal genes in PD

Substantial evidence from cell biology, genetics, and GWAS identifies the dysfunction of lysosomes as a primary cellular mechanism in PD susceptibility and pathology.^{107,108} The strong genetic evidence linking Gaucher's disease (GD, an autosomal recessive lysosomal storage disorder (LSD) caused by the loss of function of the glucocerebrosidase gene, *GBA1*) and PD risk raises the intriguing hypothesis that the LSDs and PD could share a common pathogenic mechanism. *GBA1* mutation carriers have a 5–6 fold increased risk to develop PD but display low penetrance. A study combining whole exome sequencing and exome-wide genotyping has identified ~54 LSD genes as attractive candidate risk factors for PD.⁸ Multiple lysosomal pathways and functions could be affected due to mutations in lysosomal genes culminating in PD pathology.

Ceramide metabolism

Ceramides (Cer) are a collective of bioactive lipids that belong to the class of sphingolipids composed of a sphingosine moiety and fatty acid.^{109,110} The brain has the highest proportion of lipid content and the lipid rafts in the neuronal plasma membrane are enriched in Cer.¹⁰⁹ Cer is also the backbone of glycosphingolipids (GSLs) that are synthesized by the linking of the hydrophobic ceramide moiety to the hydrophilic glycan head group and constitutes cerebrosides, globosides, and gangliosides.¹¹¹ In addition to shaping the cell membrane, Cer also acts as bioactive molecules taking part in critical cellular events. For instance, Cer accumulation induces death in dopaminergic neurons by inhibiting the PI3K/AKT survival pathway and activating BAK/BAX-mediated apoptosis.¹⁰⁹ There is a growing body of evidence that links Cer metabolism to PD. Brain and plasma ceramide levels are increased in sporadic PD patients^{112,113} (although this might not reflect the events occurring in the membrane of specific organelles). Ceramide synthase expression is increased, and ceramide acyl chain length is altered in sporadic PD.¹¹⁴ Plasma ceramide levels are correlated with cognitive deficits in PD.¹¹² In addition, several studies have highlighted that sphingolipid metabolites including glucosylceramide, psychosine, glucosylsphingosine, and gangliosides that build-up in LSDs can alter α -syn homeostasis triggering its accumulation and aggregation in neurons.¹¹⁰

Mutations in the lysosomal enzyme glucocerebrosidase (GCase encoded by the *GBA1* gene) that breaks down GlcCer to glucose and Cer cause the most common LSD Gaucher's Disease in homozygosis while increasing the lifetime risk of PD to 5-fold in heterozygosis.¹¹⁵ This increased prevalence of PD in the *GBA1* heterozygous carriers was first observed in the Ashkenazi Jewish population.¹¹⁶ Another established autosomal-dominant risk factor is the *LRRK2* gene, and mouse models deficient in *LRRK2* displayed changes in sphingolipid composition and increased Cer levels.¹¹⁷ Genetic evidence from recent GWAS studies links several PD risk genes to Cer biology. These potentially PD-causing but sparsely studied genes implicated in Cer metabolism will be discussed in the following section.

PSAP (Prosaposin). Prosaposin is a highly conserved glycoprotein whose cleavage products give rise to the sphingolipid activator proteins/SAP (saposins) A, B, C, and D.¹¹⁸ GM2A or SAP-3 is also considered a member of the SAP family.¹¹⁹ Localized primarily in the lysosomal compartments, saposins facilitate the catabolism of glycosphingolipids by lysosomal hydrolases. They act by isolating the lipid substrate from the surrounding membrane thereby orchestrating its access to degradative hydrolytic enzymes. Saposin C is shown to be required for GCase activity and its deficiency causes Gaucher's Disease.¹²⁰ Mutations in Saposin D, a lysosomal acid ceramidase (aCerase) activator, cause a rare variant type of Gaucher's Disease.^{121,122} Saposin B whose deficiency is known to cause metachromatic leukodystrophy (MLD), promotes hydrolysis of sulfatides by Arylsulfatase A (ARSA) and globotriaosylceramide (Gb3) by α -Galactosidase A (GLA),¹²³ while saposin A may cause atypical Krabbe's Disease (KD) presumably due to altered galactosylceramidase (GALC) activity.^{118,124,125} Recently studies were conducted in various cohorts and ethnic groups around the world to explore the genetic role of *PSAP* variants in PD. Oji et al.¹²⁶ were the first to provide novel genetic evidence for the contribution of saposin D mutations in PD. Three pathogenic mutations in the saposin D domain of *PSAP* were identified in patients from Japanese and Taiwanese cohorts with autosomal dominant inherited Parkinson's disease. *PSAP*-PD patient-derived iPSC neurons displayed an abnormal accumulation of autophagic vacuoles, perturbed autophagic flux, α -syn aggregation, and retention of *PSAP* in the endoplasmic reticulum.¹²⁶ While a follow-up study by Chao et al.¹²⁷ corroborated the findings of Oji et al. in a larger sample size, Zhao et al.¹¹⁸ reported but a modest effect for *PSAP* saposin D mutations in PD pathogenesis in a large Chinese mainland cohort.

Studies in saposin D^{-/-} mice showed substantial accumulation of 2-hydroxy fatty acid containing-ceramide (HFA-Cer) in the brain and kidney along

with severe renal tubular degeneration and cerebellar Purkinje neuronal degenerating displaying symptoms of progressive motor dysfunction and ataxia, the phenotypic abnormalities being evident only well into adulthood. Saposin D^{-/-} mice did not show any hallmarks of Farber's Disease (caused by deficiency of acid ceramidase) such as skin nodules or lesions, joint involvements, or hepatosplenomegaly. This might be because of partial defects in the degradation of Cer upon specific saposin D deficiency as indicated by the comparatively less HFA-Cer load from the human Farber's disease.¹²⁸ Given the rarity of *PSAP* variants, further studies need to be done in different ethnic cohorts to obtain substantial evidence to conclusively attribute the role of *PSAP* variants in PD. In addition, the molecular implication of such variants for the homeostasis of PD-relevant lysosomal pathways, especially in human iPSC-derived models, will require more work in the future.

SCARB2. The scavenger receptor class B member 2 gene (*SCARB2*) encodes the lysosome membrane protein 2 (LIMP-2) also known as CD36L2. It is a type III glycoprotein having two transmembrane domains, that traffics GCase from the ER through the Golgi and endosomes to the lysosomes in a mannose-6-phosphate independent manner relying on the di-leucine motif (L475 and I476) in cytoplasmic C-terminus.^{129–132} Investigation in COS1 cells by Kuronita et al. attributes a function for LIMP-2 in the normal biogenesis and maintenance of endosomes and lysosomes.¹³³ LIMP-2 was most recently linked to actin myoclonus renal failure (AMRF), where its deficiency causes myoclonic epilepsy and glomerulosclerosis.¹³⁰ The single nucleotide C/T polymorphism rs6812193 near the *SCARB2* gene locus and its association with sporadic PD has been widely studied in different ethnic cohorts but with inconclusive evidence of susceptibility.^{2,131,134–138} Although an association between *SCARB2* genotype at rs6812193 and an increase in PD risk was observed, this did not appear to be mediated by GCase activity according to the observations from dried blood spots (DBS) assays of PD patients.¹²⁹ Limitations of DBS assays and other non-live cell assays is that they do not recapitulate the endogenous context, where GCase needs to be correctly localized and its substrate is not exclusively available in a soluble form. In contrast, Rothaug et al. demonstrated a buildup in α -syn levels and reduced GCase expression and activity in LIMP2-deficient mice, resulting in neurotoxicity in dopaminergic neurons.¹³⁹

More studies need to be conducted to establish the importance of PD-linked *SCARB2* polymorphisms in PD-relevant models (neurons derived from patient iPSCs for instance) to have conclusive evidence for *SCARB2* as a

susceptibility factor in PD. It is not excluded that *SCARB2* may also bear other PD-relevant functions, such as regulating the sorting of lysosomal hydrolases other than GCase.

ATP10B. Compound heterozygous *ATP10B* mutation was initially identified, along with a biallelic loss-of-function mechanism in early-onset PD (EOPD) in a Belgian cohort.¹⁴⁰ This claim was challenged by Real et al. as part of the IPDGC (International Parkinson's Disease Genomics Consortium) who reported an absence of enrichment of putative compound heterozygous or homozygous carriers in patients with PD (including EOPD).¹⁴¹ Studies on a Japanese cohort also failed to identify pathogenic variants.¹⁴² Overall, more evidence is needed to conclude if *ATP10B* classifies as a PD-causing gene.

ATPase phospholipid transporting 10B is a P4-type ATPase that acts as a lipid flippase in late-endolysosomes to translocate GlcCer and phosphatidylcholine (PC) from the luminal to the cytoplasmic leaflet of the late-endolysosomal membrane.¹⁴³ P4-type ATPases including ATP10B (except for 9A and 9B) form a heterodimer with CDC50A. CDC50A is a predicted β -subunit of the flippase complex and is important to chaperone P4-type ATPases to the plasma membrane.¹⁴⁴ ATP10B seems to function in the same pathway as GCase, both regulating lysosomal GlcCer levels. ATP10B is necessary for the function of lysosomes and membrane integrity. Its knock-down increases the susceptibility of cortical neurons to cell death by dysregulated GlcCer and PC homeostasis leading to lysosomal dysfunction.¹⁴⁰

GALC. *GALC* encodes galactosylceramidase (GalCase), a lysosomal enzyme responsible for hydrolyzing galactosylceramide (GalCer) and galactosylsphingosine (GalSph).¹¹⁰ Mutations in *GALC* cause a lethal infantile disease known as Krabbe's Disease (KD) or globoid cell leukodystrophy (GLD).¹¹⁰ A correlation between some genetic variants of *GALC* and increased PD risk has been reported.^{8,66} Senkevich et al. recently reported that mutations that increase GalCase activity levels are potentially causal in PD.⁹ The SNP rs8005172 of *GALC* association with late-onset PD (LOPD) was also replicated in a small Chinese cohort.¹⁴⁵ The knockout of *GALC* did not affect monomeric α -syn accumulation in iPSC-derived Ngn2-induced neurons under basal conditions.¹⁴⁶ The exact mechanism of how enhanced GalCase activity is associated with Parkinson's disease needs to be investigated by further studying models that express variants that induce enhanced GalCase expression and activity. The impact of changes in GalCase activity on α -syn build-up and aggregation also requires investigation, to determine if some overlap with the GCase/ α -syn axis exists.

GLA. α -galactosidase A (α -Gal A) hydrolyzes terminal galactosyl residues from various complex glycoconjugated ceramides in lysosomes.¹¹⁰ *GLA* is located on the X chromosome. Its mutation and subsequent deficiency in α -Gal A activity leads to Fabry disease, characterized by the marked accumulation of glycosphingolipids including globotriaosylceramide (Gb3) and globotriaosylsphingosine (lyso-Gb3).^{110,147} A polymorphism for *GLA* within the proximal promoter region that may compromise its gene expression was previously identified in a PD patient.¹⁴⁸ Significantly low lysosomal α -Gal A activity was reported in the substantia nigra, DBS, and in leukocytes of PD patients compared to controls but no concurrent changes in Gb3 levels were observed.^{116,149} In α -Gal A-deficient mouse brains, concomitant overexpression of Gb3 was required for the development of symptoms of parkinsonism such as spontaneous tremors, delayed movements, and gait disturbances.¹⁵⁰ Nelson et al.¹⁴⁷ report a novel finding of α -Gal A deficiency in the temporal cortex of LOPD post-mortem brains that also harbors pathological α -syn accumulation. In addition, the changes in α -Gal A showed a significant correlation with GCase levels suggesting a potential cooperative role in PD.^{147,151} Further studies are warranted to prove this observation.

ARSA. Arylsulfatase is a lysosomal hydrolase that converts sulfatides, a major component of myelin, to galactosylceramides and its complete deficiency causes the autosomal recessive LSD metachromatic leukodystrophy (MLD).^{110,152} Pathogenic and protective mutations in *ARSA* were reported to be associated with PD.¹⁵³ In a recent pre-print, four out of six independent cohorts analyzed by burden analysis followed by meta-analysis showed evidence for association. In addition, a E384K variant was possibly found to segregate with PD in two families. However, the authors indicate that this will need confirmation in larger cohorts and in familial studies to ensure association.¹⁵⁴

At the molecular level, *ARSA* was reported to interact with α -syn via a non-lysosome-related molecular chaperone function. While *ARSA* gene deficiency led to higher levels of α -syn aggregation, secretion, and propagation from cell-to-cell in SH-SY5Y, *ARSA* functional mutations altered this process.¹⁵³ While the protective N352S *ARSA* variant showed a high binding affinity to α -syn and reduced α -syn aggregation, the pathogenic L300S variant had the opposite effect.¹⁵³ However, follow-up studies failed to prove the association of pathogenic L300S^{154,155} or protective N352S^{155,156} mutations with PD. A study by Li et al.¹⁵⁷ on the Chinese Han population showed an ELISA-based quantitative assessment of *ARSA* and α -syn levels in PD patients with and without cognitive impairment (PD-CI). PD-CI patients showed a low plasma *ARSA* level and high α -syn

levels which correlated with disease progression.¹⁵⁷ Given the emerging interest in lysosomal-targeted therapeutic strategies for PD, further larger case-control studies on different cohorts and ethnic groups need to be carried out to confirm the role of *ARSA* in PD.

ASAH1. Acid Ceramidase (aCerase) hydrolyzes Cer to Sph and free fatty acid. Ceramidases are classified based on their optimal pH for catalysis; aCerase (*ASAH1*), neutral Cerase (*ASAH2*), and alkaline Cerase (*ACER1*, 2, and 3).^{110,158} Deficiency in *ASAH1* causes two rare inherited disorders: Farber Disease (FD) and SMA-PME.¹⁵⁹ Due to the low life expectancy of FD/SMA-PME patients, there is a lack of correlation studies with PD although large GWAS studies showed the burden of mutation in *ASAH1*. Recent work showed that inhibition of aCerase activity in *GBA1*-PD-derived dopaminergic neurons resulted in lower levels of α -syn accumulation.¹⁶⁰ This effect could be explained as a result of Cathepsin D activity stimulation by elevated Cer levels leading to enhanced degradation of α -syn.¹¹⁰ The hypothesis that a similar mechanism could be at play as a result of mutations in other genes that cause elevated Cer levels will be important to test in the future.

SMPD1. Acid sphingomyelinase (ASMase), encoded by *SMPD1*, converts sphingomyelin to Cer and phosphorylcholine. Biallelic mutations in *SMPD1* are known to cause Niemann Pick's Disease type A or B (NPA/B).¹¹⁰ The association of *SMPD1* to PD or synucleinopathy risks have been reported in 7 independent cohorts that include ethnicities like Ashkenazi Jews,^{161–163} Chinese,^{164–166} and European.⁸ The variants L302P (the founder mutation of NPA/B) and p.fs330^{162,163} were shown to associate with an increased risk of susceptibility to PD in Ashkenazi Jews. R591C¹⁶⁴ increased PD risk in a Chinese cohort. Reduced ASMase activity was also associated with increased α -syn levels in BE(2)-M17 dopaminergic cell line and Hela cells.¹⁶² L302P and p.fsP330 mutants caused retention of ASMase in the endoplasmic reticulum, leading to the absence of the enzyme in lysosomes. At the structural level, the L302P and A487V mutations are in the catalytic domain and are likely to destabilize it and prevent proper folding.¹⁶² Given the strong similarity of *SMPD1* with *GBA1*, the most prominent risk factor for PD, causing closely related lysosomal storage disorders and having a common hydrolysis product (Cer), evidence suggests a shared pathway for both genes which requires further investigation.^{161,167}

HEXA/HEXB. β -hexosaminidase (HEX) is responsible for the metabolism of glycosphingolipid ganglioside GM2. The three isoforms of HEX result from the combination of α and β subunits encoded in *HEXA* and *HEXB*

genes,¹⁶⁸ namely: the HEX A isoform (HEX α +HEX β heterodimer), HEX B (HEX β +HEX β homodimer) and HEX S (HEX α +HEX α homodimer). The α -subunit of HEX hydrolyzes the ganglioside GM2 and requires the GM2 activator saposin peptide for its activation. Loss-of-function mutations in the *HEXA* gene cause Tay-Sachs disease, also called GM2 gangliosidosis. This lysosome storage disorder results from the aberrant hydrolysis of GM2 ganglioside. Mutations in *HEXB* cause Sandhoff disease, a variant type of Tay-Sachs disease. Tay-Sachs and Sandhoff disease are neurodegenerative diseases where the progressive loss of neurons occurs in the CNS due to the accumulation of GM2 ganglioside.¹⁶⁹

Variants in the LSD gene *HEXB* were recently confirmed as risk factors in PD patients.⁸ Deletion of *HEXA* in mice causes α -syn pathology and GM2 accumulation in the SNpc.¹⁷⁰ Deficient HEXA activity in patients affected by GM2 gangliosidosis was suggested to lead to parkinsonian-like symptoms but the sample size was too low to make a conclusive inference.¹⁷¹ Recently, a neuroprotective effect was attributed to increased HEX activity, in a *HEXB*^{-/-} Sandhoff mice model overexpressing HEXA and HEXB, reducing the degeneration of dopaminergic neurons induced by α -syn overexpression. This appeared to be mediated by restricting α -syn association to lipid compartments.¹⁷² It is hypothesized that the direct interaction between α -syn and GM2 drives α -syn accumulation. More in-depth investigations will help in better delineating the involvement of this enzyme in PD pathogenesis.

Cholesterol metabolism

In mammals, cholesterol is required to maintain the integrity, fluidity, organization, and biophysical properties of cell membranes.¹⁷³ In neurons, cholesterol is found in abundance in the synaptic lipid rafts, interacts directly with neurotransmitters, and is necessary for proper synaptic transmission and synaptogenesis.¹⁷³ The brain depends on the *de novo* synthesis of cholesterol by brain resident cells like astrocytes.

Cholesterol synthesis is a multistep, ER-localized process where the rate-limiting and regulatory step is catalyzed by 3-hydroxy-3-methylglutaryl-coenzyme-A reductase (HMGCR). Regulation of cholesterol synthesis is achieved through the binding of sterol-regulatory element-binding protein-2 (SREBP-2), a transcription factor, to the sterol-regulatory element-1 (SRE-1) in the promoter region of *HMGCR*, thereby inducing its expression and associated cholesterol biosynthesis.^{173,174} Cholesterol is delivered to neurons from astrocytes via an ApoE-cholesterol complex. The ApoE-cholesterol complex released from the astrocytes via the ATP-binding cassette transporter protein A1 (ABCA1), then binds to the low-density lipoprotein (LDL) receptors (LDLR) and the LDLR-related protein 1 (LRP1) expressed on the

neuronal membranes and is internalized into the neuron. Cholesterol is released by the hydrolysis of the complex within the late endosome/lysosome and is trafficked into subcellular compartments by the action of membrane-bound Niemann–Pick type C1 (NPC1) and luminal NPC2 proteins. This cholesterol is critical for neurons to build the membranes that constitute axons, dendrites, synapses, and synaptic vesicles.¹⁷³

NPC1/NPC2. Mutations in the *NPC1* gene result in an LSD termed Niemann Pick's Disease type C (NPC) which is characterized by the accumulation of unesterified cholesterol, glycosphingolipids, and other fatty acids in the endo-lysosomal system.¹⁷⁵ A single I1061T substitution results in misfolding of the NPC1 protein and subsequent removal by ER-associated degradation.¹⁷⁶

There are reports of disease-associated mutations in PD patients,¹⁷⁷ Parkinsonism in carriers of *NPC1* heterozygous mutants,¹⁷⁸ and a correlation between altered cholesterol metabolism and neurodegeneration.¹⁷⁹ Increased cholesterol deposition as a result of dysregulated cholesterol trafficking in neurons was reported in both NPC and PD.¹⁸⁰ Evidence supporting a link between cholesterol levels and PD includes: (i) presence of high cholesterol derivatives levels in the plasma and brain of PD patients, (ii) elevated levels of oxidized cholesterol derivatives increase α -syn levels and aggregation, (iii) decreased α -syn levels in neuronal cells upon treatment with statins, (iv) hypercholesterolemia has been linked to the higher vulnerability of DA neurons and resulting PD-like symptoms.¹⁷⁹ Controversial studies also link statin therapy, a cholesterol-lowering drug, to the risk of PD.¹⁸¹ Recently, an absence of association of a subset of NPC1 variants with PD, REM-sleep behavior, and DLB was reported in European cohorts, although a possibility exists that association might be found in other populations, or for other variants.¹⁸² The possibility of NPC variants, even though rare, playing a causal role in PD cannot be excluded.¹⁸³ The association of *NPC1/NPC2* variants with PD risk requires further elucidation and validation through widespread population genetic studies.¹⁷⁹

NPC1 is a large 13-transmembrane domain glycoprotein located in late endosomes. NPC1 possesses a 'sterol-sensing domain' formed by 5 of its transmembrane segments. This domain is found in similar proteins that are thought to sense cholesterol. NPC1 can sense rising cholesterol levels in membranes causing it to bud and be delivered to target organelles. NPC2, on the other hand, is a soluble glycoprotein that is targeted to lysosomes via the mannose 6 phosphate route. NPC2 was determined *in vitro* to employ a collisional mechanism to rapidly transport cholesterol from donor to acceptor membranes and facilitate the transfer of cholesterol to destined organelles.¹⁷⁵ Excessive accumulation of chole-

terol in lysosomes caused by NPC1/NPC2 mutations causes its transglucosylation by GCase,¹⁸⁴ leading to toxic accumulation of GlcChol and GalChol that can be hypothesized to cause neuropathology.¹⁷³ Multilamellar bodies (MLB), intracellular membrane structures of lysosomal origin composed mainly of non-degraded phospholipids and cholesterol, are a characteristic feature of *GBA1*-PD and NPC.¹⁷³ MLBs are hypothesized to facilitate aberrant interaction of α -syn and cholesterol promoting a prion-like propagation of α -syn from neuron-to-neuron.¹⁸⁰

It has been shown that alterations in the lipid composition of organelle or vesicle membranes affect their ability to fuse with target membranes. SNARE complex proteins, SNAP-23 and syntaxin 4 present at the plasma membrane and VAMP7 at the surface of lysosomes mediate their fusion, thus regulating exocytosis.¹⁸⁵ Interestingly, these proteins are localized in the cholesterol-rich lipid rafts of vesicular structures like rat brain synaptosomes.¹⁸⁶ Any dysregulation in cholesterol metabolism can affect the proper distribution of the lipids on the membranes which can impair essential lysosomal functions leading to neurodegeneration.

Glycosaminoglycans metabolism

Glycosaminoglycans (GAGs) are a family of linear and negatively charged polysaccharides with a molecular weight of approximately 10–100 kDa. GAGs are present in most cell types, within the cells, and in their surroundings.¹⁸⁷ GAGs are exquisitely complex structurally due to modifications in types of residues and their glycosidic bonds, sulfation levels, and positions, as well as chain lengths. GAGs are classified based on the hexosamine, hexose, or hexuronic acid in the disaccharide repeating building blocks and the glycosidic bonds linking these units. Accordingly, five main types of GAGs exist: nonsulfated GAGs, such as hyaluronic acid (HA), and sulfated GAGs, including heparin and heparan sulfate (HS), chondroitin sulfate (CS), dermatan sulfate (DS) and keratan sulfate (KS). The most extensively studied GAGs are heparin and HS due to their high sulfation and plethora of biological activities. In mammals, all GAGs except HA are linked to a core protein and are thus proteoglycans (PGs). Mutations in many GAG-degrading enzymes have been associated with the LSD, mucopolysaccharidosis (MPS). Burden analysis in a discovery cohort was confirmed in a meta-analysis, supporting the role of MPS genes as modifiers of *GBA1*-PD risk.¹⁰⁸ It was shown that intracellular sulfated GAGs like HS and CS regulate cathepsin D activity, the main α -synuclein degrading enzyme, and that their presence is required for α -synuclein intracellular aggregation,¹⁸⁸ accumulation,^{189–191} uptake,^{189–191} and transcellular propagation.^{189,190} Although it remains to be determined, it could be hypothesized that GAGs act in PD by modulating the activity of lysoso-

mal enzymes such as cathepsins,¹⁹² or through regulation of endo-lysosomal fusion by proteoglycans.¹⁹³ A role for HS proteoglycans as receptors of α -syn aggregates has also been established,^{189,191} possibly contributing to the prion-like spread of these proteopathic seeds and progression of PD pathogenesis as shown in worms.¹⁹⁴

NAGLU. *NAGLU* encodes N-acetylglucosaminidase, an enzyme that hydrolyses the terminal N-acetyl-D-glucosamine residues in HS. Defects in this gene cause MPS type IIIB (MPS-IIIB), also known as Sanfilippo syndrome B, an LSD characterized by accumulation of HS in lysosomes and HS urinary excretion. Mutation in the *NAGLU* gene leading to perturbed HS degradation was recently suggested to play a role in α -syn accumulation in PD, although the precise mechanism remains to be determined.¹⁹⁵ *NAGLU* expression was also found to be decreased in idiopathic PD-patient-derived dopaminergic neurons.¹⁹⁶

IDUA. *IDUA* encodes iduronidase (glycosaminoglycan alpha-L-iduronohydrolase), an enzyme that hydrolyzes terminal unsulfated alpha-L-iduronosidic linkages in DS and HS promoting the degradation of these GAGs. Iduronidase works closely and sequentially with other GAG hydrolyzing enzymes like ARSB (discussed below) thereby maintaining GAG physiological levels. *IDUA* mutations, which are inherited in an autosomal recessive fashion, result in the MPS 1 phenotype.¹⁹⁷ Defective alpha-L-iduronidase results in an accumulation of its substrates within various cell types including neurons, and fibroblasts.¹⁹⁸ No mechanistic explanation of *IDUA* involvement in PD has been described to date. It could be hypothesized that impairment of HS degradation plays a role in α -syn accumulation and prion-like spread in PD.

GUSB. *GUSB* was identified as the candidate gene of a novel PD locus.⁴ *GUSB* encodes the hydrolase β -glucuronidase, a homotetramer, which removes glucuronic acid from GAGs such as HS. Mutations in *GUSB* result in the LSD MPS-VII. *GUSB* was identified as the candidate gene of a novel PD locus.⁴ How *GUSB* is involved in PD is unknown. The impairment in HS metabolism could be hypothesized as a player in α -syn accumulation in PD.

ARSB. Arylsulfatase B belongs to a group of arylsulfatase enzymes present in lysosomes. ARSB hydrolyzes N-acetyl-D-galactosamine4-sulfate units in GAGs. ARSB mainly targets dermatan sulfate and chondroitin sulfate.¹⁹⁹ Mutated ARSB is linked with the recessive LSD MPS-VI or Maroteaux-Lamy syndrome. Four PD cases in the cohort studied by Jansen et al. were

found to carry loss-of-function variants in the *ARSB* gene that were previously linked to MPS-VI.²⁰⁰ Arylsulfatase B belongs to a group of arylsulfatase enzymes present in lysosomes. ARSB hydrolyzes N-acetyl-D-galactosamine4-sulfate units in GAGs. ARSB mainly targets dermatan sulfate and chondroitin sulfate.^{199,200}

Sialic acid metabolism

Sialic acids are a diverse family of sugars that carry negative charges. They play essential roles in normal biological functions and disease mechanisms. Sialic acids are predominantly found at the terminal positions of oligosaccharide chains of most glycans (glycoproteins and gangliosides). Sialidases, or neuraminidases, hydrolyze the α -glycosidic linkages between sialic acid residues and carbohydrate groups of glycoproteins and glycolipids.²⁰¹

NEU1. NEU1, NEU2, NEU3, and NEU4 are the four types of sialidases identified in mammals.²⁰² They display differences in their expression patterns, localizations, and functions. NEU1, which is lysosomal, is implicated in exocytosis, immune response, phagocytosis, and assembly of elastic fiber.²⁰³

The sialylation of cell surface carbohydrates in the CNS plays an essential role in the modulation of neuronal and glial behavior.²⁰⁴ The extent of sialylation is dynamic and fluctuates during brain development and also in neurological disorders.²⁰⁴ In vertebrates, polysialic acid (PSA) is one of the most biologically relevant glycotopes and is highly enriched in the neuronal cell adhesion molecule (NCAM) of the brain.²⁰⁵ NEU1 can affect neuronal migration and positioning by regulating PSA levels.²⁰⁶ NEU1 also inhibits glutamate exocytosis in hippocampal neurons by interfering with Ca^{2+} signaling.²⁰⁷ NEU1 exists as part of a heterotrimeric complex together with β -galactosidase and Cathepsin A in the lysosome.²⁰⁸ NEU1-mediated regulation of sialylation is relevant to the pathogenesis of Alzheimer's disease owing to its control over the levels of APP and its proteolytic cleavage.²⁰⁹ In a MPTP mouse model of PD, administration of GM1 ganglioside (a form of sialylated glycosphingolipid) results in significant biochemical and behavioral recovery.²¹⁰ Sialidases such as NEU1 have been proposed as an alternative to GM1, due to low bioavailability and low blood-brain barrier (BBB) penetrance of this ganglioside, to enzymatically increase the amount of endogenous GM1 in the brain by hydrolysis of polysialogangliosides.²¹⁰

SLC17A5. *SLC17A5* encodes the lysosomal membrane protein SLC17A5 (sialin). Sialin is a 12-transmembrane domain proton-coupled exporter of sialic acid (Neu5Ac) and other acidic hexoses from the lysosomes. Bi-allelic pathogenic variants in *SLC17A5* cause an extremely rare

condition called lysosomal free sialic acid storage disorder (FSASD), which is a neurodegenerative, multisystemic disease.²¹¹ *SLC12A5* gene variants can lead to loss-of-function and/or mistargeting of the transporter. *SLC17A5* variants in heterozygosity, including R39C and K136Q, have been identified as risk factors for PD.⁸ Since sialic acid is a competitive inhibitor of lysosomal neuraminidases, it is possible that excessive accumulation of free sialic acid could cause secondary storage of sialoglycoproteins and gangliosides in FSASD tissues. This could result in clinical symptoms, particularly in the CNS similar to LSDs.

Peptide metabolism

GRN. *GRN* encodes the precursor protein progranulin (PGRN) (593 amino acids, 68.5 kDa) of the granulin peptides.²¹² Cleavage of progranulin is mainly achieved by serine proteases elastase, proteinase 3, and matrix metalloproteinase to produce a variety of active 6 kDa granulin peptides named granulin A, granulin B, granulin C, etc. This processing occurs either in the extracellular matrix or the lysosomes.²¹² It was shown that progranulin trafficking to lysosomes is mediated by interaction with SORT1 (sortilin).²¹³ Progranulin shows striking similarities to prosaposin such as the existence as a pro-polypeptide, cleavage to active components and functions in the lysosome, and utilization of sortilin for trafficking. Progranulin and prosaposin were shown to influence and facilitate each other's trafficking to lysosomes via sortilin interaction.^{214,215}

A role for PGRN and granulin has been found in development, inflammation, cell proliferation, and proteostasis.²¹² The inflammatory response is mediated by the proteolytic processing of anti-inflammatory PGRN into pro-inflammatory granulin peptides.²¹⁶ A higher risk of frontotemporal dementia (FTD) was directly correlated to mutations in *GRN* that lead to reduced function or decreased levels of the protein.^{217–220} The *GRN* locus was also recently designated as contributive to the lifetime risk of numerous neurodegenerative diseases (NDDs) such as AD, PD, and ALS to different extents.^{4,221,222} A recent GWAS analysis conducted by Nalls et al. reports a first-of-its-kind strong causal functional inference that links genetic risk for these three common disorders to variations of *GRN* gene expression in the brain,²²³ suggesting a gene-level mechanistic connection that transverse a spectrum of NDDs.

Progranulin seems to be highly involved in lysosomal function and organization. It colocalizes with Lamp1, its transcription is mediated by the lysosomal biogenesis transcription factor TFEB²²⁴ and it seems to positively regulate prosaposin delivery to lysosomes and CTSD activation.^{225–227} *GRN* mutation shares similar pathology to CHMP2B deficiency,²¹² an endo-lysosomal sorting protein discussed above. Progranulin is required for the

lysosomal presence of GCase/LIMP-2 and recruits HSP70 to GCase to prevent its cytoplasmic aggregation under stress conditions in *PGRN* KO mice.²²⁸ Recently, the same team developed a clinically relevant model (PG9V mice) to study mechanistic aspects and identify potential therapeutics on GD and PD and provided convincing evidence *in vivo* and *ex vivo* on the paramount role of *PGRN* in pathologies linked to *GBA1* mutation. The PG9V mice developed neurodegenerative manifestations of neuropathic GD and PD, such as neurodegeneration, inflammation, and protein aggregation in the brain. *PGRN* was also shown to directly affect GCase lysosomal delivery and activation. A BBB penetrant *PGRN* derivative ND7 ameliorated GD manifestations and PD pathology in this mouse model.²²⁹ These results suggest that *GRN* dysfunction could lead to a defective endo-lysosomal system by affecting the delivery and activation of many lysosomal enzymes.

Cathepsins. The cathepsin family has a total of 15 known proteases. Based on their catalytic mechanism they are subgrouped into three categories: serine cathepsins (A and G), cysteine cathepsins (B, C, F, H, K, L, O, S, V, W, and X), and aspartic cathepsins (D and E).^{230,231} Most cathepsins display optimal activity at low pH, making them key players in autophagy and the endo-lysosomal system.^{230,232} Of all lysosomal proteases, the most abundant are the aspartate protease cathepsin D (CTSD) and the cysteine cathepsins B and L (CTSB, CTSL). They are especially present in the CNS.²³⁰ Many proteins associated with neurodegeneration like α -syn, β -amyloid (A β), or huntingtin (Htt) are substrates of one or more cathepsins, signifying a requirement of cathepsin-mediated degradation for neuronal homeostasis.²³⁰ GWAS studies have found CTSD⁸²³³ and CTSB⁶⁶ as modulators of PD risk.

CTSB. *CTSB* encodes the lysosomal cysteine protease known as Cathepsin B. In addition to its endopeptidase function, CTSB also has a dipeptidyl carboxypeptidase activity.²³⁴ Physiological functions of cysteine cathepsins include protein, neuropeptide, and hormone processing, major histocompatibility complex (MHC) class II-mediated antigen presentation, bone remodeling, apoptosis, and keratinocyte differentiation while also having a role in several disease pathologies like cancer, inflammation, psoriasis, muscular dystrophy, and neurodegenerative diseases.²³⁵ In the CNS, CTSB is preferentially expressed in the neocortical and hippocampal neurons. CTSB is the most abundant lysosomal protease of the papain family and is important for the physiological turnover of cell proteins in neurons.²³⁵ CTSB is a major causative factor of microglia-induced neuronal apoptosis and plays a role in the activation of the proinflammatory

caspases 1 and 11 mediated neuronal apoptosis.^{236,237}

CTSB and more efficiently CTSL derived from purified mouse brain and liver lysosomes seemed to be able to proteolyze the central and amyloid region of monomeric α -syn and thereby prevented its fibrillization *in vitro*.²³² A stimulatory effect of anionic phospholipids on cathepsin activity was also observed in the same study. However, CTSB was also shown to generate C-terminally truncated species of α -syn in symptomatic *SNCA* A53T transgenic mice.²³⁸ A protective CTSB variant rs1293298 was detected as a modifier in the onset and the risk in *GBA1*-PD and Lewy body dementia,^{7,66} probably via increased CTSB expression in the brain. Mechanistically, it appears that CTSB is mediating prosaposin processing into saposin C, which coactivates GCase in the lysosome. Thus, risk variants that decrease *CTSB* expression are thought to contribute to further hindering GCase activity in *GBA1*-associated PD. A recent study found a novel, rare CTSB G284V variant in PD patients of a Polish-German family from Poland. Gly284 is conserved in evolution and lies within the heavy chain of the enzyme and a mutation could disrupt its activity. Functional studies in patient-derived fibroblasts failed to provide any clear evidence for this mutation.²³³ Studies in patient-derived iPSC reprogrammed DA neurons or microglia might provide better insight into the role of the G284V variant in PD.

CTSD. Cathepsin D is a lysosomal aspartic protease involved in the lysosomal recycling and degradation of many substrates, including α -syn.²³⁹ Homozygous inactivation mutations in *CTSD* have been related to multiple age-related neurodegenerative diseases involving aggregation of misfolded proteins, like neuronal ceroid lipofuscinosis (NCL), AD, and PD.²⁴⁰ Genetic variants of *CTSD* were found to be associated with dysfunction of lysosomes and PD pathology in a large screening of PD patients.⁸ There is conflicting evidence on the impact of *CTSD* on α -syn clearance. While most variants in Robak et al. failed to display an effect on protein function, maturation, and localization, the PD-associated variants p.A239V had increased enzymatic activity and enhanced α -syn clearance.²⁴⁰ Studies conducted in the dopaminergic neurons of *C. elegans*, mice, and humans confirm that *CTSD* actively clears α -syn *SNCA* aggregates and restores endo-lysosomal and autophagy homeostasis.²⁴¹ Cathepsin D levels and activity were rescued in *GBA1* mutant dopaminergic neurons treated with GCase replacement or chaperone therapy, driving a decrease in monomeric α -synuclein suggesting a possible interaction between GCase and *CTSD*. Ceramide, the byproduct of GCase, can bind and activate *CTSD*.²⁴² Based on these results, lower levels of *CTSD* protein and activity seem to mediate at least in part the increased levels

of monomeric α -syn in *GBA1* mutant neurons.²⁴³ The role of CTSD in microglia-mediated α -syn clearance will be important to investigate and could represent a promising target for therapy.

ABCB9. Several polymorphisms on the *ABCB9* gene were reported where rs7953894 was found to be a significant genetic risk factor in PD.³ Interestingly, a recent study reported alterations in the methylation of CpG islands in genes from PD patients' saliva and blood samples. *ABCB9* was one of the genes that had hypomethylated CpG alterations (along with *LARS*, a mitochondrial gene, and *C1orf200*) and was associated with the SNP rs73408862.²⁴⁴

ATP-binding cassette (ABC) transporters are transmembrane proteins that actively pump endo- and xenobiotic solutes across cellular membranes against a concentration gradient using energy derived from ATP hydrolysis.²⁴⁵ ABC genes can be divided into seven subfamilies (ABC1, MDR/TAP, MRP, ALD, OABP, GCN20, and White).^{245,246} In cancer, the multidrug resistance (MDR) phenotype is attributed to the overexpression of these proteins.²⁴⁵ ATP-binding cassette sub-family B member 9 is encoded by the *ABCB9* gene and belongs to the MDR/TAP subfamily that mediates MDR and antigen presentation.^{246,247} *ABCB9* (TAPL) is localized in the lysosomes and was shown to have the ability to transport polypeptides 6–59 amino acids long from the cytosolic space into the lysosomal lumen.²⁴⁸ A molecular mechanism of *ABCB9*'s involvement in PD is lacking. Several polymorphisms on the *ABCB9* gene were reported where rs7953894 was found to be a significant genetic risk factor in PD.²⁴⁹ Interestingly, a recent study reported alterations in the methylation of CpG islands in genes from PD patients' saliva and blood samples. *ABCB9* was one of the genes that had hypomethylated CpG alterations (along with *LARS*, a mitochondrial gene, and *C1orf200*) and was associated with the SNP rs73408862.²⁴⁴

Ion channels

TMEM175. As mentioned above, *TMEM175* is part of the strongest PD disease risk loci (chromosome 4p16.3 region).⁹² Human transmembrane protein 175 (*TMEM175*) is a K⁺ channel located in late endosomes and lysosomes.⁹² *TMEM175* was recently identified as a proton-activated proton pump and is a proton 'leak' channel. *TMEM175* helps sets the optimum pH of lysosomes via a negative feedback mechanism. *TMEM175* deficiency led to over-acidity of lysosomes, reduced proteolytic activity, and α -syn aggregation in vivo.²⁵⁰ In a previous study, *TMEM175* was shown to regulate lysosomal function, autophagy, mitochondrial respiration, and α -syn aggregation in SH-SY5Y and rat hippocampal neurons.^{3,251} The lead SNP (rs34311866) that

could drive the PD risk is a missense variant, p. M393T in the 10th transmembrane domain of *TMEM175*.³ Molecular modeling indicates that this mutation is likely to destabilize the transmembrane domain of the protein, impacting its assembly or maturation.²⁵² The M393T variant caused reduced lysosomal levels of *TMEM175* and partial phenocopy of KO, including more acidic lysosomes at steady state and upon starvation, reduced GCase activity, and a trend toward decreased autophagy. Overexpression of wild-type *TMEM175*, but not the M393T variant caused a reduction of phosphorylated α -syn in vitro in rat hippocampal neurons.²⁵³ The PD-associated mutants *TMEM175* did not display impaired lysosomal localization in HeLa cells.²⁵² Additional studies are required to fully grasp the influence of *TMEM175* variants on PD risk and if other genes in this locus are implicated in PD.

ATP6V0A1. The GWAS study by Nalls et al. identified several polymorphisms in the *ATP6V0A1* gene of PD patients with rs34779517 having the most significance.²⁵⁴ *ATP6V0A1* (ATPase H⁺ Transporting V0 Subunit A1) encodes a component of vacuolar ATPase (V-ATPase), a multisubunit proton pump that mediates the acidification of eukaryotic intracellular organelles. V-ATPase consists of a transmembrane V0 domain and a cytosolic V1 domain. The GWAS study by Nalls et al. identified several polymorphisms in the *ATP6V0A1* gene of PD patients with rs34779517 having the most significance.²⁵⁴ Impaired lysosomal targeting of *ATP6V0A1* has been implicated in a neurodegenerative LSD in turn affecting lysosomal acidification.²⁵⁵ A similar mechanism could be at play in PD, although this needs to be tested.

ATP6V1D. *ATP6V1D* was also identified by GWAS analysis as a PD-linked gene.³ It encodes the V1 domain D subunit protein ATPase H⁺ transporting V1 subunit D that contains the ATP catalytic site for the transport of H⁺ to mediate the acidification of endo-lysosomes.²⁵⁶ Mutations in the gene might also affect lysosomal acidification leading to dysfunction.

Other endo-lysosomal genes

The endo-lysosomal system is a complex network of coordinated and intertwined processes, where the lysosome is the terminal point of convergence. Many proteins have overlapping functions in endosomes and lysosomes and are hence distributed in both cellular compartments. Several common endo-lysosomal genes involved in vesicle trafficking and function have been implicated as PD risk candidates.⁴

VAMP4

The protein encoded by this gene is a member of the vesicle-associated membrane protein (VAMP)/

synaptobrevin family and might function in TGN-to-endosome transport. *VAMP4* rs11578699 polymorphism was associated with PD in a large GWAS meta-analysis of idiopathic PD.⁴ Genetic interaction between an *LRRK2* variant and *VAMP4* has been suggested.²⁵⁷ It has also been suggested that *VAMP4* may be implicated in PD by regulating endocytic mechanisms. As a Golgi v-SNARE, *VAMP4* physically interacts with *LRRK2* (independently from Rab29) to regulate membrane fusion during retrograde trafficking from endosomes to Golgi.²⁵⁸ *VAMP4* also modulates synaptic vesicle retrieval in Ca^{2+} -dependent spontaneous neurotransmission, through its dileucine motif that can recruit clathrin adaptor proteins and induce its trafficking.²⁵⁹

VT11B

VIT1B encodes the vesicle transport through interaction with t-SNAREs 1B (VT11B). This protein is part of a protein complex that is required for late endosome-lysosome membrane fusion.²⁶⁰ It enables the SNARE binding activity in recycling endosomes through the interaction with t-SNAREs in the target membrane and interacts with the chloride channel CFTR in epithelial cells. VT11B depletion reduces CFTR activity by affecting its polarized trafficking to the plasma membrane.²⁶¹ VT11B is also localized in lysosomes and the Golgi, where its function remains to be elucidated. Whether and how VT11B is implicated in PD pathogenesis remains to be determined.

SPPL2B

This gene encodes the transmembrane protein signal peptide peptidase-like 2B (SPPL2B), a member of the GXGD family of aspartic proteases. SPPL2B has two conserved catalytic motifs that are localized within its membrane-spanning regions. Interestingly, SPPL2B is targeted through the secretory pathway to the endosomes, lysosomes, and the plasma membrane. It has been shown to mediate the shedding of several substrates all belonging to type 2 transmembrane protein subfamily²⁶² such as intramembrane cleavage of TNF- α (allowing the cytosolic release of its intracellular domain (ICD) that triggers interleukin-12 production²⁶³ concomitant with the liberation of TNF- α 's C-terminal part in the extracellular space,²⁶⁴ *Bri2* (Itm2b) which is genetically linked to Familial British and Danish dementia,²⁶⁵ and Fas ligand (FasL) whose ICD can negatively regulate gene transcription.²⁶⁶ There is an increasing interest in the role of TNF- α signaling in mediating dopaminergic neuron loss in PD models,²⁶⁷ but whether this is affected in any way by SPPL2B activity remains to be verified.

ZFYVE26

This gene encodes a protein, also known as SPASTIZIN, that contains a PI3P-binding FYVE zinc finger domain. This domain is suggested to target these proteins to membrane lipids through interaction with PI3P in the membrane, where it has been hypothesized to recruit Beclin 1 and other partners involved in autophagosome maturation.²⁶⁸ Loss of function mutations in SPASTIZIN has been linked to autophagosome maturation defects and reduction of autophagosome-lysosome fusion in cells from patients with autosomal recessive spastic paraplegia-15, a motor disease that presents with Parkinsonism in 16% of cases.^{269,270} Depletion of SPASTIZIN in primary hippocampal neurons also leads to the accumulation of autophagosomes.²⁶⁸ Accumulation of autophagosomes in DA neurons could lead to cell death resulting in parkinsonism.

ATP11B

The rs2270968 SNP that lies in close proximity to the *ATP11B* gene was identified to have a strong association to PD risk in a Han Chinese GWAS study.²⁷¹ *ATP11B* was also identified in *Drosophila* as a differentially expressed gene in glia compared to neurons in the aging *drosophila* brain.⁶⁴ ATP11B (ATPase Phospholipid Transporting 11B) is the catalytic component of a P4-ATPase flippase complex. The flippase transports aminophospholipids, phosphatidylserines (PS), and phosphatidylethanolamines (PE) from the outer to the inner leaflet of intracellular membranes powered by the hydrolysis of ATP.²⁷² It is also suggested to contribute to the maintenance of membrane lipid asymmetry in the endosomal compartment.²⁷² ATP11B was shown to regulate the synaptic plasticity of hippocampal neurons via the mitogen-activated protein kinase 14 (MAPK14) signaling pathway.²⁷³ For this gene as well, there is a scarcity of evidence for its molecular role in PD pathogenesis.²⁷³

CLCN3

CLCN3, which was identified by GWAS as a PD gene,³ encodes the CIC3 vesicular $2\text{Cl}^-/\text{H}^+$ exchanger. It is majorly located in the early and late endosomes and to some extent in SVs. Its main function is attributed to maintaining ion homeostasis and acidity in the endosomal lumen. Guzman et al. demonstrated a role for CIC3 in controlling the glutamatergic synaptic strength in cultured hippocampal neurons and identified excessive release of glutamate in *CICn3*^{-/-} neurons as a cause for neurodegeneration.²⁷⁴ CIC3 forms heterodimers with CIC4 which adds to the latter's stability. CIC4 was also shown to compensate for CIC3 function caused by uncoupling mutations and lacked detectable phenotype in mice as opposed to *CICn3*^{-/-275} in CA1 hippocampal pyramidal neurons. The lack of

CIC3 together with the concomitant depletion of its heteromerization partner CIC4 may impair intracellular trafficking as caused by other mutations in endo-lysosomal ion transport proteins leading to neurodegeneration pathology like PD.

Discussion and future directions

In this review, we have focused on lesser-studied PD-linked genes encoding proteins that act at the level of the endo-lysosomal compartment. Even though many endo-lysosomal genes have genetic evidence of their involvement in PD, or have been shown to affect PD pathogenesis in various animal and/or cellular models, the underlying molecular and cellular mechanisms often remain unclear. How PD neuropathology is driven by genetic variants has proven difficult to grasp for familial PD genes, and it will likely be even more challenging for risk factors. Insights into these mechanisms will be key to properly targeting these genes therapeutically. To do so, the various stakeholders need to overcome several challenges. First, the sharing of reagents (e.g. antibodies, KO or overexpression cell lines, patient-derived iPSCs, plasmids, animal models, etc...) and their associated quality control and validation data will help address the lack of tools available to study poorly characterized genes and proteins. For instance, initiatives like that of YcharOS (<https://ycharos.com/>), a public interest Canadian company that openly shares validation data of commercially available antibodies for all human proteins in applications such as Western blot, immunostaining, and IP,²⁷⁶ will help fill this gap. For cell lines, an increasing number of groups and organizations are setting up banks of cell models accessible for experimental use at a reasonable price (e.g. Parkinson's Progression Markers Initiative (PPMI), The Neuro Early Drug Discovery Unit (EDDU)). Secondly, an initiative towards studying these less-characterized genes needs to be actively encouraged. Taking on the challenge of studying poorly characterized genes and proteins can be daunting, due to the limited availability of tools and knowledge from previous literature.²⁶ Young researchers should be able to embrace the inherent risks associated with such undertakings, which might seem less profitable from an individual point of view than studying more "popular" genes and pathways. The ongoing change in mentality, where an increased variety of outputs for research impact are recognized rather than solely the number and impact factor of publications,^{277,278} will certainly help de-risk such studies at the individual level. Specific calls from funding bodies dedicated to the study of poorly characterized genes and pathways are also likely to re-equilibrate the balance. Hopefully, this will push the field to go beyond the historical bias towards a limited set of PD-linked genes,

whether they encode endo-lysosomal proteins or not.

The mounting genetic evidence for the role of genes implicated in lysosomal lipid metabolism, and the enrichment of LSD genes in PD risk factors, highlight the importance to delve deeper into the targeted analysis of subcellular compartments rather than more complex samples. To understand the impact of the dysfunction of endo-lysosomal proteins on the function of endosomes and lysosomes in an unbiased manner, analyzing their specific composition rather than that of whole cells/tissues/biofluids is a promising avenue. Previous methods relying on biochemical enrichment such as sucrose gradients can be contaminated with other organelles and require large amounts of material that are poorly compatible with analyzing human samples from PD-relevant cell types (e.g. iPSC-derived neurons, post-mortem brain tissue). In addition, they rely on time-consuming protocols that increase the chances of degradation of labile components, and loss of integrity of purified organelles. In the past years, systems have been engineered whereby expression of a lysosome-specific or endosome-specific tag allows rapid immunoprecipitation-mediated isolation of intact, highly pure lysosomes (Lyso-IP)^{279,280} or endosomes (Endo-IP)²⁸¹ from cultured cells. Another approach, specific to lysosomes, is based on the use of magnetic-plasmonic hybrid nanoparticles coupled to dextran, which can be taken up via the endocytic pathway.²⁸² Purified organelles can then be analyzed by mass spectrometry-based proteomics, lipidomics, or metabolomics, revealing alterations that may go unnoticed when analyzing more complex samples such as whole-cell lysates.^{283–285} Applying these methods in human iPSC-derived cells and organoids (possibly patient-derived) as well as animal models will likely identify key molecular changes that drive the dyshomeostasis of lysosomes and endosomes.

An unexplored area in PD genetics is the possible perturbation of non-canonical protein-coding sequences. Extensive evidence from several groups points to an underestimation of the coding potential of our genome, with thousands of proteins being expressed from open reading frames located in lncRNAs, in pseudogenes, in the 5' and 3' UTR of mRNAs, or overlapping the canonical coding sequence of cellular mRNAs but in a different reading frame.^{286–291} High-confidence experimental proof is accumulating that validates the expression of these alternative proteins (using Ribo-Seq or mass spectrometry), but also their functional importance in several key cellular processes.^{288,291–294} For instance, it is estimated that 30% of cancer-causing synonymous mutations alter unannotated ORFs,²⁹⁵ and these non-canonical ORFs can be essential for the survival of cancer cells or cell fitness.^{287,293} The vast major-

ity of alternative proteins remain uncharacterized, and it will be important for geneticists to test the possibility that PD-linked SNPs alter such non-canonical coding sequences. Even though generally smaller than canonical proteins, alternative proteins have been shown to localize to various subcellular compartments, and some likely play an important role in endo-lysosomal homeostasis.

Beyond variations in protein-coding sequences, endo-lysosomal dysfunction could arise from perturbations in other genetic elements causing transcriptional, translational, or post-translational deregulations of endo-lysosomal PD genes and proteins. Potential perturbators include non-coding RNAs, enhancers/silencers, and other transcription efficiency modulators. For instance, a *GBA1* regulation network was recently described in cultured immortalized cell lines, where the transcript of its pseudogene *GBAP* can act as a sponge for miR-122, thereby facilitating the maintenance of adequate glucocerebrosidase levels and activity.²⁹⁶ It remains to be determined whether this pathway is active in disease-relevant cell types and *in vivo*. Another miRNA, miR-128, was shown to repress autophagy by downregulating the master regulator of the autophagy-lysosome pathway TFEB, sensitizing nigral DA neurons to α -syn-induced toxicity.^{297,298} Overall, this suggests that pseudogenes and miRNAs could play an important role in PD pathogenesis.²⁹⁹ In the future, approaches such as chromosome conformation capture to study gene regulatory networks,³⁰⁰ and HITS-CLIP/CLIP-seq or other genome-wide methods to identify targets of miRNAs and other non-coding RNAs^{301,302} will likely inform on how PD-linked SNPs in non-coding and intergenic regions could affect the homeostasis of the endo-lysosomal compartment.

Genes that have not been identified as PD risk factors in genetic studies can still be of therapeutic interest for restoring proper endo-lysosomal function.²⁹⁷ While some PD-linked genes may not be easily targeted, some of their upstream or downstream functional partners may be more readily druggable. Exploring such avenues will increase the chances of drug repurposing. Only a deeper mechanistic understanding of disease processes will allow these possibilities, and increasing funds allocated to basic research will be key in progressing towards curing PD.

Finally, from a therapeutic development point of view, deciphering the mechanistic causes underlying endo-lysosomal system failure that are specific to individual patients based on their unique genetic variations, could yield better results than a one-size-fit-all approach, whether it be for familial or sporadic forms of PD. For instance, ameliorating lysosomal activity might be of limited impact if the targeting of material to degrade is inefficient due to defects in the endosomal sorting machinery. By combining the advances in precise

molecular and cellular phenotypic screening assays for endo-lysosomal function with the use of patient-specific, iPSC-derived cell types of interest (beyond just dopaminergic neurons), it might be possible to evaluate which particular disease-causing dysfunction to target therapeutically, even in the absence of known genetic liabilities. Alternatively, given the intricate interplays between multiple pathways (synaptic vesicles trafficking, endosomal sorting, lysosomal degradation, proteostasis, mitochondrial health, etc...), therapeutic advancement could also benefit from multi-pronged approaches to ameliorate endo-lysosomal system function at several levels simultaneously, to ultimately reestablish neuronal homeostasis in PD but also other diseases involving endo-lysosomes.

CRedit authorship contribution statement

Amitha Muraleedharan: Conceptualization, Writing – original draft, Writing – review & editing.
Benoît Vanderperre: Conceptualization, Writing – original draft, Writing – review & editing, Funding acquisition, Supervision.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations:

α -syn, Alpha Synuclein; AA, amino acids; ABCB9, ATP-binding cassette sub-family B member 9; aCerase, Acid Ceramidase; ARSA, Arylsulfatase A; ARSB, Arylsulfatase B; ASAH1, N-acylsphingosine amidohydrolase 1; ASMase, Acid sphingomyelinase; ATP10B, ATPase phospholipid transporting 10B; ATP11B, ATPase phospholipid transporting 11B; ATP13A2, ATPase cation transporting 13A2; ATP6V0A1, ATPase H⁺ transporting V0 subunit a1; ATP6V1D, ATPase H⁺ transporting V1 subunit D; Cer, ceramide; CFTR, Cystic fibrosis transmembrane conductance regulator; CLCN3, Chloride Voltage-Gated Channel 3; CLIP-Seq, Cross-linking and immunoprecipitation; CS, chondroitin sulfate; CTSB, Cathepsin B; CTSD, Cathepsin D; DA, dopaminergic; DGKQ, Diacylglycerol kinase theta; EOPD, Early-onset Parkinson's Disease; ESCRT, endosomal sorting complexes required for transport; FSASD, free sialic acid storage disease; GAG, Glycosaminoglycans; GAK, Cyclin G-associated kinase; Gal, galactose; GALT, Galactosylceramidase gene; GalCase, Galactosylceramidase; GalCer, galactosylceramide; GalSph, galactosylsphingosine; Gb3, globotriaosylceramide; GBA1, glucosylceramidase beta 1; GCase, β -Glucocerebrosidase; GD, Gaucher's disease; GLA, galactosidase alpha; GlcCer, glucosylceramide; GlcSph, glucosylsphingosine; GM2, GM2 ganglioside; GRN, granulin; GTP, Guanosine triphosphate; GUSB, Beta-glucuronidase; GWAS, Genome wide association studies; HEX, beta-hexosaminidase; HITS-CLIP, High-throughput sequencing of RNA isolated by crosslinking immunoprecipitation; HS, Heparan sulfate; ICD, Intracellular domain; IDUA, Alpha-L-Iduronidase; ILV, intraluminal vesicle; INPP5F, Inositol polyphosphate-5-phosphatase F; IP, Immunoprecipitation; iPSC, induced pluripotent stem cells; LacCer, lactosylceramide; LDL, Low-density lipoprotein; LIMP2, Lysosomal integral membrane protein 2; lncRNA, long non-coding RNA; LOPD, Late-onset Parkinson's Disease; LRP1, Low density lipoprotein receptor-related protein 1; LRRK2, Leucine-rich repeat kinase 2; LSD, lysosomal storage disease; miRNA, microRNA; MPS, Mucopolysaccharidoses; mRNA, messenger RNA; MVB, Multi vesicular bodies; NAGLU, N-acetylglucosaminidase, alpha; NEU1, Neuraminidase 1; NPC, Niemann-Pick's disease type C; PD, Parkinson's Disease; PG, Proteoglycans; PINK1, PTEN-induced kinase 1; PIP, phosphatidylinositol phosphate; PSAP, Prosaposin; SAP, Saposin; SCAMP, Secretory carrier-associated membrane protein; SCARB2, scavenger receptor class B

member 2; SH3GL2, SH3 domain containing GRB2 like 2; SLC17A5, solute carrier family 17 member 5; SMPD1, sphingomyelin phosphodiesterase 1; SNAP, synaptosome associated protein; SNARE, SNAP receptor; SNCA, synuclein alpha; SNP, single nucleotide polymorphism; Sph, sphingosine; SPPL2B, Signal peptide peptidase-like 2B; SV, Synaptic vesicles; SYNJ1, Synaptojanin 1; SYT, Synaptotagmin; TGN, trans-Golgi network; TMEM, Transmembrane protein; TNF-alpha, Tumor Necrosis Factor alpha; VAMP, Vesicle associated membrane proteins; VPS, Vacuolar protein sorting-associated protein; VTI1B, Vesicle transport through interaction with t-SNAREs homolog 1B; ZFYVE26, Zinc finger, FYVE domain containing 26

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